Linear-Time Ensemble Clustering with Adaptive Self-Refinement of Clusters

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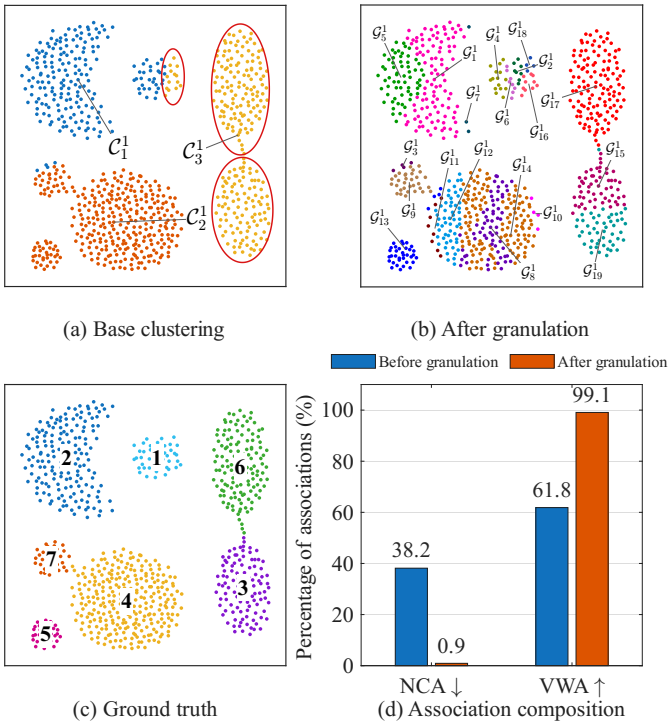
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***Abstract*—Ensemble clustering combines multiple base clusterings to obtain a robust consensus partition. Although dynamic anchor-based methods improve scalability, representative methods treat each original base cluster as an indivisible structural unit. This coarse granularity becomes problematic when a low-purity base cluster contains multiple high-purity sub-clusters, as valid within-sub-cluster associations coexist with noisy cross-sub-cluster associations, making it difficult to distinguish reliable from noisy associations. To solve this problem, we propose Linear-Time Ensemble Clustering with Adaptive Self-Refinement of Clusters (TEAR). Without accessing raw data features, TEAR first constructs consensus pseudo-features from base clusterings and adaptively refines base clusters into finer-grained structural units called granules. However, structural refinement alone may be insufficient for global consensus learning, as samples from the same underlying class may reside in different granules without direct co-clustering associations in the ensemble. To address this challenge, TEAR then integrates granulated hypergraph reconstruction and high-order self-expression regularization into dynamic anchor learning, exploiting indirect structural relationships to obtain a global consensus representation and extract the final consensus partition. Dedicated acceleration strategies further enable TEAR to achieve linear-time complexity with respect to the number of samples. Experiments on 10 datasets against 18 representative state-of-the-art methods demonstrate superior overall clustering performance and competitive scalability. The source code and supplemental material are available at https://github.com/Haocai-Liang/TEAR.**

***Index Terms*—Ensemble clustering, adaptive cluster granulation, dynamic anchor, high-order connectivity, scalable clustering.**



(a) Base clustering (b) After granulation

(c) Ground truth (d) Association composition

Fig. 1. Effect of granulation on a base clustering generated from the Aggregation[1] dataset. (a) Base clustering with three clusters, where the yellow cluster $\mathcal{C}_3^1$ contains three high-purity sub-clusters enclosed by red ellipses. (b) Granulation reveals fine-grained and high-purity sub-clusters. (c) Ground truth. (d) Granulation increases the proportion of valid within-sub-cluster associations and reduces the proportion of noisy cross-sub-cluster associations.

I. Introduction

Ensemble clustering integrates the partitions produced by multiple base clusterings to obtain a more robust consensus partition [1]–[4]. Since it typically relies solely on the results of base clusterings without accessing raw data features, it is particularly valuable in real-world applications with data privacy requirements [5]. Many state-of-the-art (SOTA) ensemble clustering methods rely on the co-association (CA) matrix to capture pairwise sample relationships [6]–[9]. But constructing and processing the $n \times n$ CA matrix incurs substantial computational and memory costs, limiting their scalability to large datasets [10], [11]. To improve scalability, anchor-based ensemble clustering methods learn compact sample–anchor representations for consensus partitioning [12], [13]. Among them, dynamic anchor-based methods further enhance flexibility and adaptability by jointly optimizing the anchors and sample–anchor representations [14], [15].

Despite their strengths, these methods still rely on the original base clusters, whose reliability may constrain subsequent consensus learning. As illustrated in Fig. 1(a) and (c), the low-purity base cluster C_3^1 comprises several high-purity sub-clusters corresponding to distinct ground-truth classes. Such a base cluster contains both valid within-sub-cluster associations (VWAs) and noisy cross-sub-cluster associations (NCAs). In

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[1]https://cs.joensuu.fi/sipu/datasets/

representative dynamic anchor-based methods [16], each base cluster is encoded as a hyperedge connecting all its members and treated as an indivisible structural unit, making it difficult to distinguish VWAs from NCAs.

To mitigate the adverse effects of low-purity base clusters on consensus learning, previous studies have introduced cluster-level weighting strategies based on validity measures such as entropy [17], [18]. Nevertheless, a single cluster weight can only adjust the overall contribution of a cluster and cannot characterize the heterogeneous reliability of sample associations within it. For a low-purity cluster containing multiple high-purity sub-clusters, assigning a high weight may reinforce NCAs, whereas reducing its weight weakens VWAs. This limitation cannot be resolved merely by improving cluster quality estimation, as it stems from the restricted flexibility in representing heterogeneous sample associations when base clusters are treated as indivisible structural units. To overcome this structural granularity bottleneck, it is necessary to adaptively split base clusters into more reliable fine-grained structural units, thereby laying the foundation for subsequent global consensus learning.

Specifically, realizing such adaptive structural refinement and global consensus learning presents three key challenges: First, ensemble clustering typically operates solely on base clustering results, without access to raw features or sample distances for identifying sub-cluster boundaries. Thus, reliable structural evidence must be extracted directly from the ensemble. Second, different base clusters require different levels of refinement. Indiscriminate splitting may disrupt reliable associations, while a fixed refinement granularity may fail to capture heterogeneous internal structures. Therefore, the refinement process must adaptively determine to what extent each base cluster should be split. Third, refining local structures alone may be insufficient to establish a global consensus representation, as samples from the same underlying class may be distributed across different granules and lack direct co-clustering associations in the ensemble. Although high-order connectivity can capture indirect relationships through intermediate samples [19], [20], explicitly constructing and manipulating dense sample–sample connectivity matrices incurs substantial computational and memory costs [21], [22].

To this end, we propose Linear-**T**ime **E**nsemble Clustering with **A**daptive Self-**R**efinement of Clusters (**TEAR**). First, TEAR constructs consensus pseudo-features from base clusterings and adaptively refines base clusters using a decision criterion driven by co-clustering support, aiming to suppress NCAs while preserving VWAs, as shown in Fig. 1(b) and (d). Next, using the resulting granules as a common structural foundation, TEAR integrates granulated hypergraph reconstruction and high-order self-expression regularization into dynamic anchor learning to obtain a global consensus representation, from which the final consensus partition is extracted. Finally, dedicated acceleration strategies enable TEAR to achieve linear-time complexity with respect to the number of samples.

The main contributions of this paper are summarized as follows:

- We propose a novel ensemble-driven granulation mechanism that requires no raw features. By leveraging consensus pseudo-features and a co-clustering support-based criterion, it adaptively transforms base clusters into finer-grained structural units.
- We integrate the granulated hypergraph with high-order self-expression regularization within a dynamic anchor framework, exploiting indirect structural evidence to learn a global consensus representation.
- We develop cluster-level counting and low-rank factorization strategies to accelerate adaptive granulation and alternating optimization, respectively, avoiding explicit sample-pair enumeration and large-scale matrix inversion while achieving overall linear-time complexity with respect to the number of samples.
- Experiments on 10 datasets against 18 representative SOTA methods demonstrate superior overall clustering performance and competitive scalability of TEAR.

II. Related Work

In this section, we first define ensemble clustering and review dynamic anchor-based methods. Next, we discuss locally weighted ensemble clustering. Finally, we summarize high-order connectivity guided by random walks. The main notations used in this paper are listed in Table I.

A. Ensemble Clustering and Dynamic Anchor-Based Method

Given a dataset $\mathcal{X} = \{x_1, x_2, \ldots, x_n\}$ of n samples, ensemble clustering aims to integrate a set of weak base clusterings $\Pi = \{\pi^1, \pi^2, \ldots, \pi^v\}$ to produce a final consensus partition $\mathcal{C}$ that is more robust and accurate than individual base clustering. Specifically, the p-th base clustering $\pi^p = \{\mathcal{C}_1^p, \mathcal{C}_2^p, \ldots, \mathcal{C}_{s_p}^p\}$ partitions the dataset into s_p disjoint base clusters, yielding a total of $s = \sum_{p=1}^{v} s_p$ base clusters in ensembles. Ensemble clustering operates solely on decision-level results without accessing raw data features.

Recently, dynamic anchor-based ensemble clustering methods have made significant progress [14]–[16]. By formulating all base clusterings into a hypergraph $\mathbf{H} \in \{0,1\}^{s\times n}$, these methods dynamically reconstruct it to learn the anchors and the sample-anchor similarity:

$$\min_{\mathbf{Q},\mathbf{A},\mathbf{Y}} \ \|\mathbf{H} - \mathbf{Q}\mathbf{A}\mathbf{Y}\|_F^2$$
$$\text{s.t. } \mathbf{Q}^\top\mathbf{Q} = \mathbf{I}_d,\ \mathbf{A}^\top\mathbf{A} = \mathbf{I}_l,\ \mathbf{Y} \geq \mathbf{0},\ \mathbf{1}_l^\top\mathbf{Y} = \mathbf{1}_n^\top,$$

where $\mathbf{Q} \in \mathbb{R}^{s\times d}$ serves as a projection matrix to filter out redundant information from the hypergraph, $\mathbf{A} \in \mathbb{R}^{d\times l}$ represents the anchor matrix, and $\mathbf{Y} \in \mathbb{R}^{l\times n}$ denotes the sample-anchor similarity matrix. Although dynamic anchor-based methods exhibit remarkable scalability, they universally treat base clusters (hyperedges) as indivisible units, which inherently restricts the flexibility of hypergraph reconstruction.

B. Locally Weighted Ensemble Clustering

In ensemble clustering, base clusters often exhibit diverse quality. Thus, quantifying cluster uncertainty is essential for boosting clustering performance.

TABLE I
MAIN NOTATION

Symbol	Description
n, v	Numbers of samples and base clusterings
c	Number of consensus clusters
d, l	Anchor-space dimension and number of anchors
s_p, s	Numbers of base clusters in π^p and in the ensemble
g_p, g	Numbers of granules in π^p and in the ensemble
r_p	Comparison-space dimension, $r_p = \max(g_p, d)$
o	Maximum order of high-order connectivity
$\mathcal{X}, \Pi$	Dataset and ensemble of base clusterings
π^p	The p-th base clustering
$\mathcal{C}_i^p, \mathcal{G}_i^p$	The i-th base cluster and granule of π^p
$\mathbf{H}^{(p)}$	Indicator matrix of π^p
$\mathbf{X}^{(p)}$	Granule indicator matrix of π^p
$\mathbf{U}^{(p)}, \mathcal{U}$	Granule-to-comparison-space mapping and its set
$\mathbf{V}^{(p)}, \mathcal{V}$	Anchor-to-comparison-space mapping and its set
$\mathbf{W}^{(p)}, \mathcal{W}$	Relative anchor-to-granule mapping and its set
$\mathbf{A}$	Anchor matrix
$\mathbf{Y}$	Sample–anchor representation matrix
$\mathbf{P}^t$	High-order connectivity matrix

Huang et al. [17] measured cluster uncertainty via an entropic criterion by measuring how sample distributions within a cluster are dispersed across other base clusterings, and subsequently derived the ensemble-driven cluster index (ECI) to locally weight base clusters.

Furthermore, Zhang et al. [18] observed that smaller clusters generally possess tighter connections within them. Based on this observation, they introduced normalized ensemble entropy. However, existing approaches only evaluate weights at the cluster level. A low-purity base cluster may contain multiple high-purity sub-clusters, resulting in the coexistence of valid within-sub-cluster associations (VWAs) and noisy cross-sub-cluster associations (NCAs). A cluster-level weight therefore cannot adequately characterize the heterogeneous reliability of sample associations within such a base cluster.

C. High-Order Connectivity Guided by Random Walk

To recover missing links, leveraging random walks to model high-order connectivity has emerged as a representative paradigm.

Huang et al. [19] defined a Markov transition probability matrix on a microcluster similarity graph. By executing multi-step random walks along graph paths, it accumulates multi-step probability trajectories and evaluates indirect relationships via the similarity between trajectories originating from different nodes, effectively mitigating the sparsity of the first-order co-association matrix. Furthermore, Huang et al. [20] scaled random walks to the base-cluster level to substantially reduce computational complexity. Despite their success in uncovering missing links, existing random walk-based methods still suffer from notable limitations. Random walks conducted at the microcluster or cluster level only offer approximated pairwise associations at the sample level. Besides, directly executing random walks at the sample level entails explicit matrix chain multiplications of size $n \times n$ [22], [23], making it expensive for large-scale datasets.

III. THE PROPOSED METHOD

We propose Linear-**T**ime **E**nsemble Clustering with **A**daptive Self-**R**efinement of Clusters (**TEAR**) to address NCAs and use indirect structural relationships to support global consensus learning.

A. Adaptive Self-Refinement of Clusters via Granulation

To refine the internal structure of base clusters, we construct consensus pseudo-features solely from base clustering results to guide granulation and develop an ensemble-driven criterion to adaptively determine the number of granules.

For each base clustering, we first construct pseudo-features by aggregating support from the other base clusterings in the ensemble. Let $\mathbf{H}^{(p)} \in \{0,1\}^{s_p \times n}$ denote the indicator matrix of the p-th base clustering, where s_p is its number of base clusters and the r-th row indicates membership in $\mathcal{C}_r^p$. Let $s = \sum_{p=1}^{v} s_p$ denote the total number of base clusters. By vertically concatenating the indicator matrices of all base clusterings except the p-th one, we construct the external base-clustering indicator matrix $\mathbf{R}^{(p)} \in \{0,1\}^{(s-s_p)\times n}$:

$$\mathbf{R}^{(p)} = \left[\mathbf{H}^{(1)\top}, \ldots, \mathbf{H}^{(p-1)\top}, \mathbf{H}^{(p+1)\top}, \ldots, \mathbf{H}^{(v)\top}\right]^\top.$$

The external support matrix $\mathbf{F}^{(p)}$ is constructed from $\mathbf{R}^{(p)}$ and row-wise normalized to obtain the consensus pseudo-feature matrix $\mathbf{Z}^{(p)}$:

$$\mathbf{F}^{(p)} = \mathbf{R}^{(p)\top}(\mathbf{R}^{(p)}\mathbf{H}^{(p)\top}), \quad \mathbf{Z}_{a,:}^{(p)} = \frac{\mathbf{F}_{a,:}^{(p)}}{\|\mathbf{F}_{a,:}^{(p)}\|_2}. \tag{1}$$

Specifically, $\mathbf{R}^{(p)}\mathbf{H}^{(p)\top} \in \mathbb{R}^{(s-s_p)\times s_p}$ records the number of shared samples between each cluster in the other base clusterings and each cluster in the p-th base clustering. We then left-multiply this count matrix by $\mathbf{R}^{(p)\top}$. For each sample x_a, this operation selects the clusters containing x_a in the other base clusterings and adds their overlap counts with each base cluster in π^p. Thus, $F_{a,r}^{(p)}$ measures the total number of times x_a is grouped with members of $\mathcal{C}_r^p$ across the external base clusterings, reflecting their external co-clustering support. Row-wise normalization converts these support values into relative support strengths. For any base cluster or intermediate granule $\mathcal{G}$, we apply k-means ($k = 2$) to the consensus pseudo-feature submatrix $\mathbf{Z}_{\mathcal{G},:}^{(p)}$ to produce two nonempty candidate sub-granules $\mathcal{G}_1$ and $\mathcal{G}_2$.

With the pseudo-features available to guide granulation, we next develop an external-support-based criterion to determine whether each candidate granulation should be accepted, thereby adaptively controlling the degree of refinement. We define the external co-clustering support matrix as

$$\boldsymbol{\rho}^{(p)} = \frac{1}{v-1}\mathbf{R}^{(p)\top}\mathbf{R}^{(p)},$$

where $\rho_{a,b}^{(p)}$ is the proportion of the other base clusterings that assign samples x_a and x_b to the same cluster. Thus, $\rho_{a,b}^{(p)}$ and $1 - \rho_{a,b}^{(p)}$ can be interpreted as the similarity and dissimilarity between the two samples.

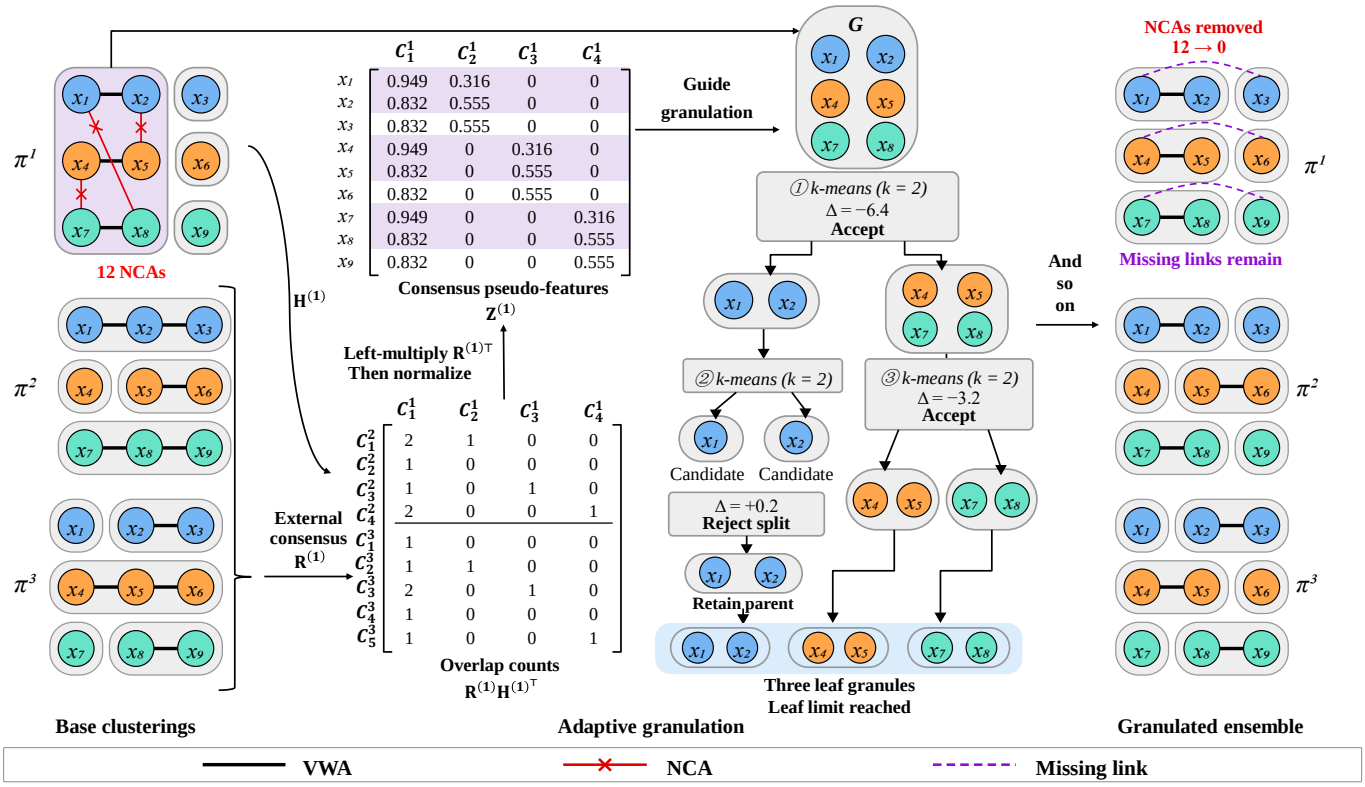


Fig. 2. Illustration of adaptive cluster granulation. Node colors indicate ground-truth classes, and gray contours denote clusters or granules.

Let $\mathcal{P}$ denote a partition of $\mathcal{G}$, with $\mathcal{P}_1 = \{\mathcal{G}\}$ representing no granulation and $\mathcal{P}_2 = \{\mathcal{G}_1, \mathcal{G}_2\}$ representing the candidate granulation. For a candidate granulation $\mathcal{P}$, we assign an inconsistency cost of $\tau[1 - \rho_{a,b}^{(p)}]$ to retaining a sample pair within the same granule and $(2-\tau)\rho_{a,b}^{(p)}$ to separating them into different granules, where $\tau \in [0, 2]$ controls the trade-off between the two costs. For each $x_a \in \mathcal{G}$, let z_a denote the index of the sub-granule in $\mathcal{P}$ that contains x_a. We define the total connection inconsistency cost as:

$$\mathcal{L}(\mathcal{P}) = \sum_{\substack{a<b \\ x_a, x_b \in \mathcal{G}}} \Big[\tau\big[1-\rho_{a,b}^{(p)}\big]\mathbf{1}(z_a = z_b) + (2-\tau)\rho_{a,b}^{(p)}\mathbf{1}(z_a \neq z_b)\Big].$$

Relative to the non-granulating baseline $\mathcal{P}_1 = \{\mathcal{G}\}$, the candidate granulation $\mathcal{P}_2 = \{\mathcal{G}_1, \mathcal{G}_2\}$ only alters the connection states of sample pairs crossing the sub-granules, while connections within each sub-granule remain unchanged and cancel out. The resulting cost change is derived as:

$$\Delta = \mathcal{L}(\mathcal{P}_2) - \mathcal{L}(\mathcal{P}_1) = \sum_{a:\, x_a \in \mathcal{G}_1} \sum_{b:\, x_b \in \mathcal{G}_2} \big[2\rho_{a,b}^{(p)} - \tau\big]. \quad (2)$$

We accept the granulation if and only if $\Delta < 0$, which is equivalent to the average external co-clustering support rate for cross-sub-granule sample pairs falling below $\tau/2$. A smaller τ leads to more conservative granulation, whereas a larger τ leads to more aggressive granulation. Each accepted granulation strictly decreases the connection inconsistency cost of the current granule $\mathcal{G}$.

Using this criterion, we recursively granulate [24], [25] the accepted sub-granules in a breadth-first manner. The current granule is retained if it contains only one sample, all normalized feature vectors in $\mathcal{G}$ are identical, or the candidate granulation is rejected. Each original base cluster produces at most c leaf granules; subsequent granulation within that base cluster terminates upon reaching this limit to prevent individual base clusters from being over-granulated.

Let $\{\mathcal{G}_i^p\}_{i=1}^{g_p}$ denote the final granules obtained for the p-th base clustering. The corresponding granulated base-clustering indicator matrix is given by:

$$X_{i,a}^{(p)} = \mathbf{1}(x_a \in \mathcal{G}_i^p), \ \mathbf{X}^{(p)} \in \{0,1\}^{g_p \times n}.$$

And the overall granulated base-clustering incidence matrix is constructed as:

$$\mathbf{X} = [\mathbf{X}^{(1)\top}, \ldots, \mathbf{X}^{(v)\top}]^\top \in \{0,1\}^{g \times n},$$

where $g = \sum_{p=1}^{v} g_p$ is the total number of granules across all base clusterings.

Fig. 2 illustrates pseudo-feature construction and recursive granulation, including accepted and rejected candidate splits.

B. Global Consensus Learning from Granules

As illustrated in Fig. 2, adaptive granulation separates samples from different classes within an unreliable base cluster, but missing links between the resulting granules and other clusters remain. Granulation may also sever some VWAs and

introduce additional missing links. We therefore build high-order connectivity upon the granulated base clusterings to capture indirect relationships among samples and guide the recovery of these missing links.

To capture high-order connectivity, we first construct a first-order connection matrix from the granule incidence matrix $\mathbf{X}$ defined in Section III-A:

$$\mathbf{P}_0 = \mathbf{D}_0^{-1/2}\mathbf{X}^\top\mathbf{X}\mathbf{D}_0^{-1/2} \in \mathbb{R}^{n\times n},$$

where $\mathbf{D}_0 = \mathrm{diag}(\mathbf{X}^\top\mathbf{X}\mathbf{1}_n)$ is the degree matrix.

To improve the quality of this first-order connection matrix, we account for differences in granule reliability using normalized ensemble entropy (NEE) [18]. For a granule $\mathcal{G}$ originating from the p-th base clustering, the proportion of its samples belonging to cluster $\mathcal{C}_r^j$ in the j-th base clustering is

$$\eta_j(r;\mathcal{G}) = \frac{|\mathcal{G}\cap\mathcal{C}_r^j|}{|\mathcal{G}|}.$$

Its aggregate NEE across all v base clusterings is defined as

$$e(\mathcal{G}) = -\frac{\sum_{j=1}^{v}\sum_r \eta_j(r;\mathcal{G})\log_2\eta_j(r;\mathcal{G})}{\log_2 s_p},$$

where s_p is the number of clusters in the p-th base clustering. We assign larger weights to granules with lower uncertainty:

$$w(\mathcal{G}) = \exp\left(-\frac{e(\mathcal{G})}{\theta v}\right),\ \mathbf{\Omega} = \mathrm{diag}\left(w(\mathcal{G}_1),\ldots,w(\mathcal{G}_g)\right), \quad (3)$$

where $\theta > 0$ controls the sensitivity of the weights to granule uncertainty. Using these weights, we define the degree matrix $\mathbf{D} = \mathrm{diag}(\mathbf{X}^\top\mathbf{\Omega}\mathbf{X}\mathbf{1}_n)$ and construct the weighted first-order connection matrix

$$\mathbf{P} = \mathbf{D}^{-1/2}\mathbf{X}^\top\mathbf{\Omega}\mathbf{X}\mathbf{D}^{-1/2} \in \mathbb{R}^{n\times n}. \quad (4)$$

The high-order connection matrices are computed as

$$\mathbf{P}^0 = \mathbf{I}_n, \quad \mathbf{P}^t = \mathbf{P}\mathbf{P}^{t-1}(t\geq 1),$$

whose entries aggregate products of connection weights along paths of length t. Through intermediate samples, these paths capture indirect relationships between samples that lack direct connections. We jointly exploit the structures of orders $t = 1,\ldots,o$ to provide information for recovering missing links.

To incorporate high-order connectivity into scalable consensus learning, we combine it with a dynamic anchor learning framework. We first align the granulated base clusterings in a common space with shared dynamic anchors:

$$\min_{\mathcal{U},\mathcal{V},\mathbf{A},\mathbf{Y}} \sum_{p=1}^{v}\|\mathbf{U}^{(p)}\mathbf{X}^{(p)} - \mathbf{V}^{(p)}\mathbf{A}\mathbf{Y}\|_F^2$$
$$\text{s.t.}\quad \mathbf{U}^{(p)\top}\mathbf{U}^{(p)} = \mathbf{I}_{g_p},\ \mathbf{V}^{(p)\top}\mathbf{V}^{(p)} = \mathbf{I}_d,\ \mathbf{A}^\top\mathbf{A} = \mathbf{I}_l,$$

where $\mathbf{A}\in\mathbb{R}^{d\times l}$ is the shared anchor matrix, and $\mathbf{Y}\in\mathbb{R}^{l\times n}$ is the sample–anchor representation matrix. Unlike the nonnegative similarity matrix in Section II-A, $\mathbf{Y}$ is an unconstrained representation matrix, allowing flexible sample representations in the shared anchor space. Their product $\mathbf{A}\mathbf{Y}$ represents all samples through the shared anchors. Since base clusterings contain different numbers of granules, their indicator matrices cannot be directly aligned with a shared representation. To learn a unified consensus, we introduce $\mathbf{U}^{(p)}\in\mathbb{R}^{r_p\times g_p}$ and $\mathbf{V}^{(p)}\in\mathbb{R}^{r_p\times d}$ to map $\mathbf{X}^{(p)}$ and $\mathbf{A}\mathbf{Y}$ into a common comparison space for each base clustering, where $r_p = \max(g_p, d)$. We denote the mapping collections by $\mathcal{U} = \{\mathbf{U}^{(p)}\}_{p=1}^v$ and $\mathcal{V} = \{\mathbf{V}^{(p)}\}_{p=1}^v$.

After aligning the base clusterings with shared anchors, we incorporate high-order self-expression [26], [27] into the sample representations to obtain the unified objective:

$$\min_{\mathcal{U},\mathcal{V},\mathbf{A},\mathbf{Y}} \sum_{p=1}^{v}\|\mathbf{U}^{(p)}\mathbf{X}^{(p)} - \mathbf{V}^{(p)}\mathbf{A}\mathbf{Y}\|_F^2 + \lambda\sum_{t=1}^{o}\|\mathbf{Y}-\mathbf{Y}\mathbf{P}^t\|_F^2$$
$$\text{s.t.}\quad \mathbf{U}^{(p)\top}\mathbf{U}^{(p)} = \mathbf{I}_{g_p},\ \mathbf{V}^{(p)\top}\mathbf{V}^{(p)} = \mathbf{I}_d,\ \mathbf{A}^\top\mathbf{A} = \mathbf{I}_l, \quad (5)$$

where $\lambda > 0$ controls the strength of high-order self-expression. The regularization term encourages $\mathbf{Y}\approx\mathbf{Y}\mathbf{P}^t$ across orders $t = 1,\ldots,o$, using $\mathbf{P}^t$ as the self-expression coefficients. Let $\mathbf{y}_a$ denote the a-th column of $\mathbf{Y}$, and let $p_{b,a}^{(t)}$ denote the (b,a)-th entry of $\mathbf{P}^t$. The t-th-order self-expression encourages

$$\mathbf{y}_a \approx \sum_{b=1}^{n} p_{b,a}^{(t)}\mathbf{y}_b,$$

where $p_{b,a}^{(t)}$ weights the contribution of sample x_b according to its t-th-order connectivity with x_a. Even when two samples have no direct connection, indirect paths can yield a nonzero weight, allowing their representations to inform each other and supporting the recovery of missing links.

C. Alternating Optimization

We solve Eq. (5) using block coordinate descent (BCD) [28], updating each variable block while keeping the others fixed.

1) Update $\mathbf{U}^{(p)}, \mathbf{V}^{(p)}$: With $\mathbf{A}$ and $\mathbf{Y}$ fixed, we jointly optimize $\mathbf{U}^{(p)}$ and $\mathbf{V}^{(p)}$ through their relative mapping:

$$\mathbf{W}^{(p)} = \mathbf{U}^{(p)\top}\mathbf{V}^{(p)} \in \mathbb{R}^{g_p\times d},$$

which allows us to optimize a single relative mapping instead of updating $\mathbf{U}^{(p)}$ and $\mathbf{V}^{(p)}$ separately. Let $\mathcal{W} = \{\mathbf{W}^{(p)}\}_{p=1}^v$ denote the set of relative mapping matrices across all base clusterings. When $g_p \geq d$, $\mathbf{U}^{(p)}$ is square orthogonal, so $\mathbf{W}^{(p)\top}\mathbf{W}^{(p)} = \mathbf{I}_d$. When $g_p < d$, $\mathbf{V}^{(p)}$ is square orthogonal, giving $\mathbf{W}^{(p)}\mathbf{W}^{(p)\top} = \mathbf{I}_{g_p}$. Thus,

$$\begin{cases}\mathbf{W}^{(p)\top}\mathbf{W}^{(p)} = \mathbf{I}_d, & g_p \geq d,\\ \mathbf{W}^{(p)}\mathbf{W}^{(p)\top} = \mathbf{I}_{g_p}, & g_p < d.\end{cases} \quad (6)$$

Using the column orthogonality of $\mathbf{U}^{(p)}$, $\mathbf{V}^{(p)}$, and $\mathbf{A}$, the reconstruction term becomes

$$\|\mathbf{U}^{(p)}\mathbf{X}^{(p)} - \mathbf{V}^{(p)}\mathbf{A}\mathbf{Y}\|_F^2 = \|\mathbf{X}^{(p)}\|_F^2 + \|\mathbf{Y}\|_F^2 - 2\,\mathrm{tr}(\mathbf{X}^{(p)\top}\mathbf{W}^{(p)}\mathbf{A}\mathbf{Y}).$$

Algorithm 1 The Proposed TEAR Algorithm

Require: Multiple base clustering results Π; number of clusters c; hyperparameters τ, λ.
Ensure: Consensus partition $\mathcal{C}$.

Precomputation: Obtain $\mathbf{X}$ by granulation; compute $\mathbf{\Omega}$ via Eq. (3), $\mathbf{B}, \mathbf{T}$ via Eq. (15), and $\mathbf{K}$ via Eq. (16).
Initialize orthogonal $\mathbf{A}, \mathcal{W}$ satisfying Eq. (6), $\mathbf{Y} = \mathbf{1}_{l\times n}/l$.
while not converged **do**
  for $p = 1, \ldots, v$ **do**
    Update $\mathbf{U}^{(p)}, \mathbf{V}^{(p)}$ through $\mathbf{W}^{(p)}$ via Eq. (8);
  end for
  Compute $\mathbf{S}$ via Eq. (9) and update $\mathbf{A}$ via Eq. (10);
  Compute $\mathbf{E} = \mathbf{A}^\top \mathbf{S}$ and update $\mathbf{Y}$ via Eq. (18);
  Check stopping criteria:
  $\max(\max_{1\le p\le v} \sigma(\mathbf{W}^{(p)}), \sigma(\mathbf{A}), \sigma(\mathbf{Y})) \le 10^{-2}$ with $\sigma(\mathbf{N}) = \|\mathbf{N}_{i+1} - \mathbf{N}_i\|_F^2 / \|\mathbf{N}_i\|_F^2$;
end while
Extract the consensus partition $\mathcal{C}$ from $\mathbf{Y}$.

Dropping the constant $\sum_{p=1}^{v} \|\mathbf{X}^{(p)}\|_F^2 = vn$, we equivalently rewrite Eq. (5) as

$$\min_{\mathcal{W},\mathbf{A},\mathbf{Y}} v\|\mathbf{Y}\|_F^2 - 2\sum_{p=1}^{v} \operatorname{tr}(\mathbf{X}^{(p)\top}\mathbf{W}^{(p)}\mathbf{A}\mathbf{Y}) + \lambda\sum_{t=1}^{o}\|\mathbf{Y} - \mathbf{Y}\mathbf{P}^t\|_F^2 \quad \text{s.t. Eq. (6)},\ \mathbf{A}^\top\mathbf{A} = \mathbf{I}_l. \tag{7}$$

The corresponding subproblem for $\mathbf{W}^{(p)}$ is

$$\min_{\mathbf{W}^{(p)}} \ -2\operatorname{tr}(\mathbf{X}^{(p)\top}\mathbf{W}^{(p)}\mathbf{A}\mathbf{Y}).$$

The subproblem is equivalent to:

$$\max_{\mathbf{W}^{(p)}} \ \operatorname{tr}(\mathbf{C}^{(p)\top}\mathbf{W}^{(p)}) \quad \text{s.t. Eq. (6)},$$

where $\mathbf{C}^{(p)\top} = \mathbf{A}\mathbf{Y}\mathbf{X}^{(p)\top}$. Let $\mathbf{C}^{(p)} = \mathbf{U}_{\mathbf{C}^{(p)}}\mathbf{\Sigma}_{\mathbf{C}^{(p)}}\mathbf{V}_{\mathbf{C}^{(p)}}^\top$ be its SVD. An optimal update is

$$\mathbf{W}^{(p)} = \mathbf{U}_{\mathbf{C}^{(p)}}\mathbf{V}_{\mathbf{C}^{(p)}}^\top. \tag{8}$$

An optimal pair of mappings is recovered from the updated $\mathbf{W}^{(p)}$ as $(\mathbf{U}^{(p)}, \mathbf{V}^{(p)}) = (\mathbf{I}_{g_p}, \mathbf{W}^{(p)})$ when $g_p \ge d$, or $(\mathbf{W}^{(p)\top}, \mathbf{I}_d)$ otherwise.

2) Update $\mathbf{A}$*:* With $\mathcal{U}$, $\mathcal{V}$ and $\mathbf{Y}$ fixed, the subproblem for $\mathbf{A}$ is

$$\min_{\mathbf{A}} \ -2\sum_{p=1}^{v}\operatorname{tr}(\mathbf{X}^{(p)\top}\mathbf{W}^{(p)}\mathbf{A}\mathbf{Y}).$$

The subproblem is equivalent to:

$$\max \ \operatorname{tr}(\mathbf{M}^\top\mathbf{A}) \quad \text{s.t.} \quad \mathbf{A}^\top\mathbf{A} = \mathbf{I}_l.$$

where

$$\mathbf{M} = \mathbf{S}\mathbf{Y}^\top, \quad \mathbf{S} = \sum_{p=1}^{v}\mathbf{W}^{(p)\top}\mathbf{X}^{(p)} \in \mathbb{R}^{d\times n}. \tag{9}$$

Let $\mathbf{M} = \mathbf{U}_\mathbf{M}\mathbf{\Sigma}_\mathbf{M}\mathbf{V}_\mathbf{M}^\top$ be its SVD. An optimal update is

$$\mathbf{A} = \mathbf{U}_\mathbf{M}\mathbf{V}_\mathbf{M}^\top, \tag{10}$$

3) Update $\mathbf{Y}$*:* With $\mathcal{U}$, $\mathcal{V}$ and $\mathbf{A}$ fixed, the subproblem for $\mathbf{Y}$ is

$$\min_{\mathbf{Y}} \ v\|\mathbf{Y}\|_F^2 - 2\sum_{p=1}^{v}\operatorname{tr}(\mathbf{X}^{(p)\top}\mathbf{W}^{(p)}\mathbf{A}\mathbf{Y}) + \lambda\sum_{t=1}^{o}\|\mathbf{Y} - \mathbf{Y}\mathbf{P}^t\|_F^2.$$

Let $\mathbf{E} = \mathbf{A}^\top\mathbf{S} \in \mathbb{R}^{l\times n}$. Denoting the convex subproblem objective by $f(\mathbf{Y})$, its gradient is

$$\nabla_\mathbf{Y} f = 2v\mathbf{Y} - 2\mathbf{E} + 2\lambda\sum_{t=1}^{o}(\mathbf{Y} - \mathbf{Y}\mathbf{P}^t)(\mathbf{I}_n - \mathbf{P}^t)^\top.$$

Setting $\nabla_\mathbf{Y} f = 0$ gives

$$\mathbf{Y}\mathbf{J} = \mathbf{E}, \tag{11}$$

where

$$\mathbf{J} = v\mathbf{I}_n + \lambda\sum_{t=1}^{o}(\mathbf{I}_n - \mathbf{P}^t)(\mathbf{I}_n - \mathbf{P}^t)^\top. \tag{12}$$

For the subsequent low-rank factorization, define

$$\mathbf{G} = \sum_{t=1}^{o}(\mathbf{I}_n - \mathbf{P}^t)(\mathbf{I}_n - \mathbf{P}^t)^\top,$$

so that $\mathbf{J} = v\mathbf{I}_n + \lambda\mathbf{G}$. Since $\mathbf{G} \succeq 0$, $v > 0$, and $\lambda > 0$, $\mathbf{J}$ is positive definite. The unique optimal solution is therefore

$$\mathbf{Y} = \mathbf{E}\mathbf{J}^{-1}. \tag{13}$$

For label extraction, we first apply stable column-wise Softmax to $\mathbf{Y}$ to obtain $\widehat{\mathbf{Y}}$. We then extract the leading c left singular vectors of $\widehat{\mathbf{Y}}^\top$, normalize their rows to unit ℓ_2 norm, and apply k-means to obtain $\mathcal{C}$. Algorithm 1 summarizes the complete procedure.

D. Computational Acceleration

We accelerate granulation and representation learning through cluster-level counting and low-rank computation, respectively.

1) Accelerating Granulation: The consensus pseudo-features in Eq. (1) are computed without forming an $n \times n$ sample similarity matrix. We first compute the sparse cross-cluster count matrix $\mathbf{R}^{(p)}\mathbf{H}^{(p)\top}$ and then left-multiply it by $\mathbf{R}^{(p)\top}$.

We also evaluate the granulation criterion using cluster-level counting without enumerating sample pairs. For a candidate granulation $\{\mathcal{G}_1, \mathcal{G}_2\}$, Eq. (2) can be evaluated as

$$\Delta = \frac{2}{v-1}\sum_{j\ne p} b_j - \tau m, \tag{14}$$

where $m = |\mathcal{G}_1||\mathcal{G}_2|$ is the number of sample pairs crossing the two sub-granules, and b_j counts those assigned to the same cluster by π^j. For each base clustering π^j with $j \ne p$, we count how many samples from each sub-granule belong to each cluster, obtaining two frequency vectors $\mathbf{q}_1$ and $\mathbf{q}_2$ with matching cluster indices. Their inner product gives $b_j = \mathbf{q}_1^\top\mathbf{q}_2$,

since multiplying the two counts for each cluster gives the number of cross-sub-granule pairs assigned to that cluster, and summing these products yields the total count. Only nonzero counts need to be stored. Constructing these vectors and evaluating the criterion across all other base clusterings requires $\mathcal{O}(v|\mathcal{G}|)$ time per candidate granulation. Further details and pseudocode are provided in the supplemental material.

2) Accelerating Optimization: Directly evaluating Eq. (13) requires inverting the $n \times n$ matrix $\mathbf{J}$, which incurs $\mathcal{O}(n^3)$ time complexity. We next exploit the low-rank factorization of $\mathbf{P}$ and apply the Woodbury matrix identity [29] to compute the update efficiently. The weighted first-order connection matrix in Eq. (4) admits the exact factorization $\mathbf{P} = \mathbf{B}\mathbf{B}^\top$, where

$$\mathbf{B} = \mathbf{D}^{-1/2}\mathbf{X}^\top\mathbf{\Omega}^{1/2} \in \mathbb{R}^{n\times g}, \quad \mathbf{T} = \mathbf{B}^\top\mathbf{B} \in \mathbb{R}^{g\times g}. \quad (15)$$

It follows that $\mathbf{P}^t = \mathbf{B}\mathbf{T}^{t-1}\mathbf{B}^\top$. Using the symmetry of $\mathbf{P}$, we rewrite the regularization matrix as

$$\mathbf{G} = o\mathbf{I}_n + \mathbf{B}\mathbf{K}\mathbf{B}^\top, \quad \mathbf{K} = \sum_{t=1}^{o}(-2\mathbf{T}^{t-1} + \mathbf{T}^{2t-1}). \quad (16)$$

Substituting this expression into Eq. (12) gives $\mathbf{J} = \alpha\mathbf{I}_n + \lambda\mathbf{B}\mathbf{K}\mathbf{B}^\top$, where $\alpha = v + \lambda o > 0$. We then apply the product form of the Woodbury matrix identity to obtain

$$\mathbf{J}^{-1} = \frac{\mathbf{I}_n}{\alpha} - \frac{\lambda}{\alpha}\mathbf{B}(\alpha\mathbf{I}_g + \lambda\mathbf{K}\mathbf{T})^{-1}\mathbf{K}\mathbf{B}^\top. \quad (17)$$

Substituting Eq. (17) into Eq. (13) yields the efficient update

$$\mathbf{Y} = \frac{\mathbf{E}}{\alpha} - \frac{\lambda}{\alpha}(\mathbf{E}\mathbf{B})(\alpha\mathbf{I}_g + \lambda\mathbf{K}\mathbf{T})^{-1}\mathbf{K}\mathbf{B}^\top. \quad (18)$$

The Woodbury identity replaces the inversion of the $n \times n$ matrix $\mathbf{J}$ with solving a $g \times g$ linear system, reducing the cubic cost from $\mathcal{O}(n^3)$ to $\mathcal{O}(g^3)$. The $g \times g$ matrix $\alpha\mathbf{I}_g + \lambda\mathbf{K}\mathbf{T}$ is factorized once and reused across iterations. The remaining products are evaluated from left to right, avoiding explicit construction of any $n \times n$ matrix.

E. Complexity Analysis

In this subsection, we analyze the time and space complexity of Algorithm 1.

1) Time Complexity: Let i_{gran} and i_{km} denote upper bounds on the total k-means iterations for each candidate granulation and label extraction, respectively. Constructing the consensus pseudo-features and generating and evaluating candidate granulations costs $\mathcal{O}(nvs + nc(i_{\text{gran}}s + v^2))$. Granule weighting and constructing $\mathbf{B}$ and $\mathbf{T}$ require $\mathcal{O}(nv^2+g^2)$. Computing $\mathbf{K}$ and precomputing $(\alpha\mathbf{I}_g + \lambda\mathbf{K}\mathbf{T})^{-1}\mathbf{K}$ together cost $\mathcal{O}(og^3)$.

Using the relative mappings, each alternating iteration updates $\mathcal{W}$, $\mathbf{A}$, and $\mathbf{Y}$. Since $l \le d$ and $\min(g_p, d) \le d$, its cost is bounded by $T_{\text{alt}} = \mathcal{O}\big(nd(v + l) + gd^2 + dl^2 + lg^2\big)$. For i alternating iterations, the optimization costs iT_{alt}. Label extraction requires $\mathcal{O}(nl^2 + i_{\text{km}}nc^2)$. In practice, $v, s, c, d, l, g, o, i_{\text{gran}}, i_{\text{km}} \ll n$. The overall time complexity is

$$\mathcal{O}\Big(n\big[s(v + c) + cv^2 + d(v + l) + c^2\big]\Big).$$

TABLE II
DETAILS OF DIFFERENT DATASETS.

No.	Dataset	#Sample	#Feature	#Class
D1	Zoo	101	16	7
D2	Breast (Diagnostic)	569	30	2
D3	Breast Cancer Wisconsin	699	9	2
D4	Wireless Indoor Localization	2,000	7	4
D5	Sensor_Readings_2	5,456	2	4
D6	Optdigits	5,620	64	10
D7	Statlog (Landsat Satellite)	6,435	36	6
D8	PD	10,992	16	10
D9	HTRU2	17,898	8	2
D10	Letter Recognition	20,000	16	26

2) Space Complexity: The consensus pseudo-features require $\mathcal{O}(ns)$ space, while the sparse indicator matrices and $\mathbf{B}$ require $\mathcal{O}(nv)$. The relative mappings and optimization variables require $\mathcal{O}(gd+dl+nl+nd)$, and the small matrices used for high-order computation and the Woodbury update require $\mathcal{O}(g^2)$. The overall space complexity is therefore

$$\mathcal{O}\big(n(s + v + l + d)\big).$$

F. Convergence Analysis

Let $\mathcal{F}$ denote the reformulated objective in Eq. (7). Let $\mathcal{W}_i = \{\mathbf{W}_i^{(p)}\}_{p=1}^{v}$ and $\mathcal{F}_i^{(0)} = \mathcal{F}(\mathcal{W}_i, \mathbf{A}_i, \mathbf{Y}_i)$ denote the mapping set and objective value at the start of iteration i. Let $\mathcal{F}_i^{(1)}$, $\mathcal{F}_i^{(2)}$, and $\mathcal{F}_i^{(3)}$ denote the objective values after updating $\mathcal{W}$, $\mathbf{A}$, and $\mathbf{Y}$, respectively, so that $\mathcal{F}_i^{(3)} = \mathcal{F}_{i+1}^{(0)}$. Each update in Eqs. (8), (10) and (18) globally minimizes its corresponding block subproblem. Therefore,

$$\mathcal{F}_{i+1}^{(0)} = \mathcal{F}_i^{(3)} \le \mathcal{F}_i^{(2)} \le \mathcal{F}_i^{(1)} \le \mathcal{F}_i^{(0)}.$$

Moreover, the reformulated objective in Eq. (7) equals that in Eq. (5) minus the constant vn. The original objective is a sum of nonnegative squared Frobenius-norm terms with nonnegative weights, so the reformulated objective is bounded below by $-vn$. The objective sequence is nonincreasing and therefore converges to a finite value.

IV. EXPERIMENT

A. Experimental Setup

We conduct experiments on 10 datasets spanning a range of sample sizes, feature dimensions, and numbers of classes. Table II summarizes their basic information, with further dataset details provided in the supplementary material. For each dataset, we run k-means with the number of clusters s_p randomly selected from $[2, \sqrt{n}]$ to generate 100 candidate base clusterings. We then randomly select 20 base clusterings for each ensemble, repeat this sampling process 10 times, and report the mean and standard deviation of the results.

Clustering quality is evaluated using Accuracy (ACC), Cohen's Kappa (Kappa), Normalized Mutual Information (NMI), Adjusted Rand Index (ARI), Purity (PUR), and F-score.

TABLE III

THE CLUSTERING PERFORMANCES OF DIFFERENT ALGORITHMS ARE MEASURED BY ACC(%). BEST RESULTS ARE HIGHLIGHTED IN **BOLD**, AND SECOND-BEST RESULTS ARE MARKED WITH AN UNDERLINE.

Method	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	Avg.
k-means (best)	87.13 ± 0.00	92.79 ± 0.00	95.57 ± 0.00	94.25 ± 0.00	60.87 ± 0.00	87.85 ± 0.00	58.60 ± 0.00	73.10 ± 0.00	91.85 ± 0.00	26.98 ± 0.00	76.90 ± 0.00
k-means (average)	67.79 ± 10.60	34.61 ± 17.65	46.04 ± 16.77	33.44 ± 21.49	25.11 ± 12.53	39.80 ± 18.69	30.16 ± 12.32	36.29 ± 17.36	9.49 ± 11.99	20.85 ± 2.81	34.36 ± 14.22
PTA-AL [19]	70.59 ± 7.40	85.15 ± 5.59	74.33 ± 15.90	91.79 ± 7.75	56.85 ± 13.64	87.11 ± 5.55	68.32 ± 3.66	75.74 ± 4.60	90.67 ± 2.93	27.16 ± 2.33	72.77 ± 6.93
PTA-CL [19]	66.93 ± 8.73	63.85 ± 4.92	64.22 ± 5.36	71.69 ± 8.90	49.47 ± 7.76	53.64 ± 12.60	23.96 ± 0.22	50.14 ± 6.29	90.84 ± 0.00	4.50 ± 0.42	53.92 ± 5.52
PTGP [19]	69.50 ± 6.89	92.34 ± 4.60	96.09 ± 0.80	93.41 ± 5.76	59.88 ± 9.27	90.01 ± 3.95	64.87 ± 5.36	76.59 ± 5.33	89.13 ± 0.49	25.58 ± 3.67	75.74 ± 4.61
SEC [30]	62.67 ± 11.66	69.44 ± 11.49	74.54 ± 8.52	59.38 ± 14.67	51.71 ± 3.91	55.81 ± 6.94	48.00 ± 8.75	44.80 ± 9.73	91.48 ± 6.22	21.63 ± 1.97	57.94 ± 8.39
RSEC-Z [31]	74.75 ± 7.47	92.64 ± 0.75	92.25 ± 7.44	95.95 ± 0.67	47.23 ± 5.33	89.08 ± 2.59	60.32 ± 5.37	–	–	–	–
RSEC-H [31]	61.88 ± 9.90	90.65 ± 5.81	72.79 ± 13.39	96.41 ± 0.70	50.03 ± 7.53	86.00 ± 4.25	55.29 ± 5.00	–	–	–	–
GoT [8]	74.26 ± 5.16	87.77 ± 6.32	95.81 ± 1.04	70.76 ± 12.21	51.46 ± 8.03	71.51 ± 9.69	70.59 ± 0.84	66.33 ± 6.24	90.27 ± 0.38	26.45 ± 1.30	70.52 ± 5.12
SPCE [32]	76.04 ± 6.06	92.74 ± 0.83	96.44 ± 0.53	87.70 ± 12.38	56.10 ± 8.08	–	–	–	–	–	–
AWEC-H [22]	71.29 ± 4.57	89.03 ± 2.60	93.05 ± 7.18	95.28 ± 1.40	–	–	–	–	–	–	–
AWEC-S [22]	73.07 ± 5.17	92.37 ± 1.13	94.55 ± 7.69	94.07 ± 6.00	–	–	–	–	–	–	–
CEAM [33]	64.85 ± 4.94	73.60 ± 14.94	94.45 ± 1.84	95.17 ± 2.07	59.57 ± 7.28	73.62 ± 6.92	50.68 ± 5.69	61.31 ± 4.32	–	–	–
FCE [34]	64.46 ± 4.26	70.32 ± 10.25	58.41 ± 2.85	72.01 ± 9.99	40.12 ± 2.67	86.60 ± 6.16	44.61 ± 4.53	66.97 ± 4.88	50.06 ± 0.00	24.77 ± 1.58	57.83 ± 4.72
YACHT [14]	76.73 ± 2.57	90.67 ± 2.22	94.31 ± 7.01	92.30 ± 5.78	50.83 ± 5.92	83.97 ± 7.19	67.95 ± 5.79	71.58 ± 5.11	58.14 ± 5.97	28.31 ± 1.24	71.48 ± 4.88
SDGCA [18]	77.52 ± 2.24	91.88 ± 3.42	96.29 ± 0.88	96.67 ± 1.27	48.74 ± 9.93	–	–	–	–	–	–
FSEC [13]	53.17 ± 10.59	88.40 ± 5.57	87.07 ± 4.68	87.97 ± 12.94	55.25 ± 5.26	84.91 ± 8.32	69.42 ± 7.22	80.15 ± 6.06	86.38 ± 2.06	28.25 ± 1.04	72.10 ± 6.37
GPEC [35]	76.73 ± 2.39	90.63 ± 3.94	85.04 ± 15.70	96.74 ± 0.53	54.42 ± 5.44	–	60.38 ± 5.85	–	–	–	–
RANGE [16]	70.99 ± 7.63	91.44 ± 12.42	94.16 ± 11.36	96.22 ± 8.71	51.40 ± 3.80	89.31 ± 3.86	65.06 ± 6.20	81.56 ± 3.93	61.87 ± 3.14	27.98 ± 1.64	73.00 ± 6.27
DREAM [15]	76.83 ± 6.16	93.41 ± 1.05	96.05 ± 0.53	96.25 ± 3.48	51.75 ± 6.33	88.68 ± 4.48	68.71 ± 7.01	77.52 ± 3.80	66.46 ± 4.44	28.21 ± 0.86	74.39 ± 3.81
TEAR (Ours)	**79.90 ± 1.87**	**95.59 ± 0.42**	**96.74 ± 0.37**	**97.32 ± 0.11**	**72.27 ± 3.34**	**91.52 ± 2.14**	**72.24 ± 4.05**	**85.38 ± 1.14**	**91.60 ± 1.90**	**29.11 ± 0.77**	**81.17 ± 1.61**

"–" indicates excessive runtime (mean execution time over 10 runs $>$ 7200 s).

TABLE IV

THE CLUSTERING PERFORMANCES OF DIFFERENT ALGORITHMS ARE MEASURED BY KAPPA(%). BEST RESULTS ARE HIGHLIGHTED IN **BOLD**, AND SECOND-BEST RESULTS ARE MARKED WITH AN UNDERLINE.

Method	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	Avg.
k-means (best)	83.37 ± 0.00	84.24 ± 0.00	90.10 ± 0.00	92.33 ± 0.00	38.83 ± 0.00	86.50 ± 0.00	52.63 ± 0.00	70.49 ± 0.00	59.19 ± 0.00	24.53 ± 0.00	68.22 ± 0.00
k-means (average)	56.95 ± 16.98	21.05 ± 15.18	27.00 ± 16.56	28.36 ± 20.80	17.58 ± 8.18	37.29 ± 18.60	25.47 ± 10.77	33.60 ± 16.98	2.31 ± 4.91	19.40 ± 2.69	26.90 ± 13.16
PTA-AL [19]	63.16 ± 8.45	65.69 ± 13.44	53.92 ± 27.06	89.06 ± 10.33	34.51 ± 23.44	85.68 ± 6.16	61.43 ± 4.63	73.03 ± 5.11	57.57 ± 7.72	24.26 ± 2.42	60.83 ± 10.88
PTA-CL [19]	59.28 ± 9.56	8.38 ± 16.08	32.75 ± 14.74	62.25 ± 11.87	17.38 ± 13.13	48.47 ± 14.03	0.20 ± 0.29	44.44 ± 7.03	-0.01 ± 0.00	0.48 ± 0.46	27.36 ± 8.72
PTGP [19]	62.12 ± 7.69	82.77 ± 11.18	91.39 ± 1.85	91.21 ± 7.69	39.70 ± 13.07	88.90 ± 4.39	57.74 ± 6.44	73.99 ± 5.93	54.06 ± 1.07	22.61 ± 3.81	66.45 ± 6.31
SEC [30]	53.01 ± 14.03	33.68 ± 24.58	30.46 ± 25.44	45.83 ± 19.56	25.17 ± 7.30	50.91 ± 7.70	35.30 ± 12.19	38.60 ± 10.90	39.16 ± 31.22	18.50 ± 2.03	37.06 ± 15.50
RSEC-Z [31]	68.64 ± 8.40	83.91 ± 1.64	83.54 ± 13.41	94.59 ± 0.89	26.40 ± 7.57	87.86 ± 2.88	52.39 ± 6.25	–	–	–	–
RSEC-H [31]	51.78 ± 13.86	80.57 ± 10.67	50.52 ± 23.54	95.22 ± 0.93	32.32 ± 8.04	84.44 ± 4.73	46.33 ± 5.90	–	–	–	–
GoT [8]	67.51 ± 5.88	72.36 ± 15.00	90.66 ± 2.42	61.01 ± 16.28	22.28 ± 14.02	68.34 ± 10.77	64.47 ± 1.01	62.59 ± 6.92	53.31 ± 1.65	23.52 ± 1.35	58.61 ± 7.53
SPCE [32]	68.14 ± 8.34	84.10 ± 1.88	92.12 ± 1.23	83.61 ± 16.51	30.59 ± 14.24	–	–	–	–	–	–
AWEC-H [22]	63.87 ± 5.74	77.00 ± 4.59	85.84 ± 13.31	93.70 ± 1.86	–	–	–	–	–	–	–
AWEC-S [22]	65.80 ± 6.61	83.74 ± 2.37	88.97 ± 14.02	92.10 ± 8.00	–	–	–	–	–	–	–
CEAM [33]	57.88 ± 5.37	44.35 ± 32.04	87.81 ± 4.10	93.56 ± 2.76	41.08 ± 8.54	70.69 ± 7.69	40.18 ± 6.90	57.01 ± 4.82	–	–	–
FCE [34]	57.25 ± 4.86	40.66 ± 20.50	16.84 ± 5.72	62.68 ± 13.32	20.16 ± 3.56	85.11 ± 6.85	33.53 ± 5.43	63.30 ± 5.43	0.12 ± 0.00	21.76 ± 1.64	40.14 ± 6.73
YACHT [14]	70.79 ± 3.18	80.14 ± 4.45	88.13 ± 13.27	89.73 ± 7.71	32.63 ± 7.10	82.19 ± 7.99	61.53 ± 6.65	68.41 ± 5.67	-12.05 ± 9.59	25.44 ± 1.29	58.70 ± 6.69
SDGCA [18]	71.39 ± 2.80	81.98 ± 8.12	91.84 ± 2.06	95.56 ± 1.70	16.56 ± 16.35	–	–	–	–	–	–
FSEC [13]	43.69 ± 12.33	73.56 ± 13.86	73.70 ± 8.71	83.95 ± 17.26	38.74 ± 6.64	83.24 ± 9.25	62.93 ± 8.57	77.94 ± 6.74	48.04 ± 4.07	25.38 ± 1.08	61.12 ± 8.85
GPEC [35]	70.52 ± 2.96	80.49 ± 7.19	72.30 ± 27.31	95.65 ± 0.71	37.11 ± 7.35	–	52.40 ± 7.00	–	–	–	–
RANGE [16]	64.47 ± 8.49	81.93 ± 24.23	88.27 ± 20.50	94.96 ± 11.62	33.88 ± 4.42	88.12 ± 4.29	58.03 ± 7.07	79.51 ± 4.37	10.23 ± 6.28	25.10 ± 1.71	62.45 ± 9.30
DREAM [15]	69.84 ± 8.23	85.74 ± 2.22	91.23 ± 1.22	95.01 ± 4.64	35.87 ± 9.48	87.42 ± 4.97	62.53 ± 8.17	75.02 ± 4.23	-12.15 ± 3.38	25.34 ± 0.89	61.58 ± 4.74
TEAR (Ours)	**74.65 ± 2.34**	**90.44 ± 0.90**	**92.92 ± 0.78**	**96.42 ± 0.14**	**59.02 ± 4.87**	**90.58 ± 2.37**	**65.94 ± 5.15**	**83.75 ± 1.26**	**60.36 ± 6.58**	**26.27 ± 0.80**	**74.04 ± 2.52**

"–" indicates excessive runtime (mean execution time over 10 runs $>$ 7200 s).

Higher values indicate better clustering performance for all six metrics.

We compare TEAR with 18 state-of-the-art (SOTA) ensemble clustering methods, comprising eight scalability-oriented methods and ten others. The former include PTA-AL, PTA-CL, and PTGP [19], SEC [30], FSEC [13], YACHT [14], RANGE [16], and DREAM [15]; the latter include RSEC-H and RSEC-Z [31], GoT [8], SPCE [32], AWEC-H and AWEC-S [22], CEAM [33], FCE [34], SDGCA [18], and GPEC [35]. Details are provided in the supplementary material. For fairness, we tune the baselines following their original papers and report their best results. For TEAR, optimization stops when the relative change falls below 10^{-2} or after 50 iterations. We fix $o = 3$, $\theta = 0.1$, and $d = l = c$, while searching $\lambda \in \{10^2, 10^3, \ldots, 10^7\}$ and $\tau \in \{0.1, 0.2, \ldots, 0.8\}$.

B. Clustering Performance Comparison

Tables III, IV, and V report the ACC, Kappa, and NMI results of TEAR and 18 compared methods on 10 datasets as means and standard deviations over 10 independent runs. ARI, PUR, and F-score results are provided in the supplementary

TABLE V

THE CLUSTERING PERFORMANCES OF DIFFERENT ALGORITHMS ARE MEASURED BY NMI(%). BEST RESULTS ARE HIGHLIGHTED IN **BOLD**, AND SECOND-BEST RESULTS ARE MARKED WITH AN UNDERLINE.

Method	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	Avg.
k-means (best)	82.90 ± 0.00	62.31 ± 0.00	72.31 ± 0.00	84.51 ± 0.00	49.88 ± 0.00	78.99 ± 0.00	61.09 ± 0.00	73.53 ± 0.00	33.94 ± 0.00	53.01 ± 0.00	65.25 ± 0.00
k-means (average)	64.24 ± 15.93	36.17 ± 6.02	44.47 ± 9.55	60.07 ± 8.79	42.10 ± 3.54	69.10 ± 7.45	50.63 ± 5.24	65.59 ± 7.78	11.47 ± 3.22	45.16 ± 8.29	48.90 ± 7.58
PTA-AL [19]	70.96 ± 4.01	46.56 ± 14.70	41.32 ± 25.06	86.49 ± 3.98	33.48 ± 20.15	84.46 ± 2.32	60.74 ± 2.11	78.65 ± 1.76	33.88 ± 6.20	38.48 ± 1.59	57.50 ± 8.19
PTA-CL [19]	69.12 ± 4.40	6.16 ± 8.32	23.05 ± 9.25	74.42 ± 10.21	19.86 ± 12.58	66.56 ± 10.56	0.45 ± 0.49	63.93 ± 6.24	0.00 ± 0.00	1.87 ± 1.93	32.54 ± 6.40
PTGP [19]	72.40 ± 4.18	64.65 ± 11.46	75.53 ± 4.53	87.26 ± 3.04	39.92 ± 12.33	85.30 ± 1.68	60.23 ± 2.95	78.46 ± 2.64	32.06 ± 0.81	37.00 ± 3.13	63.28 ± 4.67
SEC [30]	63.92 ± 8.83	23.05 ± 14.85	20.57 ± 19.66	63.31 ± 13.34	24.07 ± 7.97	65.42 ± 4.36	38.77 ± 12.67	53.99 ± 7.77	31.02 ± 20.66	30.70 ± 2.45	41.48 ± 11.26
RSEC-Z [31]	76.08 ± 3.31	61.73 ± 3.17	64.49 ± 14.24	88.52 ± 1.25	33.51 ± 4.56	82.10 ± 1.91	56.14 ± 5.52	–	–	–	–
RSEC-H [31]	60.61 ± 7.91	59.95 ± 12.41	36.42 ± 24.17	89.08 ± 1.21	35.74 ± 7.97	80.62 ± 1.90	50.71 ± 5.21	–	–	–	–
GoT [8]	74.46 ± 3.08	51.85 ± 17.03	74.01 ± 5.12	76.53 ± 6.03	21.27 ± 14.27	74.99 ± 5.87	61.88 ± 0.69	69.16 ± 4.03	28.36 ± 1.68	38.46 ± 2.05	57.10 ± 5.99
SPCE [32]	70.94 ± 6.39	62.37 ± 3.00	76.63 ± 2.76	82.96 ± 7.59	30.54 ± 9.13	–	–	–	–	–	–
AWEC-H [22]	73.19 ± 2.07	52.66 ± 4.90	71.12 ± 15.30	87.71 ± 1.86	–	–	–	–	–	–	–
AWEC-S [22]	73.47 ± 4.19	60.80 ± 4.46	75.74 ± 14.82	88.02 ± 3.31	–	–	–	–	–	–	–
CEAM [33]	71.62 ± 3.42	24.49 ± 25.31	68.77 ± 9.28	87.33 ± 3.70	38.74 ± 5.11	74.64 ± 4.67	44.30 ± 6.55	65.76 ± 2.72	–	–	–
FCE [34]	70.88 ± 2.85	17.84 ± 15.75	2.62 ± 1.41	53.94 ± 12.30	14.10 ± 4.51	79.80 ± 4.89	25.45 ± 3.22	58.76 ± 2.60	0.00 ± 0.00	31.10 ± 0.93	35.45 ± 4.85
YACHT [14]	75.11 ± 2.28	54.47 ± 6.78	72.00 ± 16.18	83.52 ± 6.78	38.97 ± 5.55	79.43 ± 3.51	58.07 ± 4.36	69.72 ± 2.09	6.52 ± 2.06	37.47 ± 0.87	57.53 ± 5.05
SDGCA [18]	75.71 ± 2.03	61.27 ± 9.41	77.08 ± 4.89	89.87 ± 1.74	27.70 ± 11.20	–	–	–	–	–	–
FSEC [13]	59.30 ± 8.75	53.67 ± 12.92	52.39 ± 9.10	83.51 ± 5.40	41.79 ± 2.86	82.62 ± 3.78	60.66 ± 3.09	78.04 ± 3.42	27.86 ± 2.73	**40.55 ± 1.19**	58.04 ± 5.32
GPEC [35]	76.75 ± 2.02	56.67 ± 6.96	58.57 ± 26.23	89.74 ± 1.10	44.04 ± 4.10	–	51.01 ± 4.02	–	–	–	–
RANGE [16]	74.91 ± 3.37	57.98 ± 22.16	74.70 ± 22.10	89.04 ± 10.79	40.58 ± 5.52	83.58 ± 1.90	61.40 ± 4.18	75.27 ± 2.07	6.19 ± 3.54	38.21 ± 1.44	60.19 ± 7.71
DREAM [15]	73.72 ± 7.43	64.04 ± 4.09	74.65 ± 2.68	88.30 ± 5.00	40.09 ± 5.00	81.58 ± 2.14	61.19 ± 3.79	71.14 ± 1.40	3.90 ± 0.63	36.89 ± 0.83	59.55 ± 3.30
TEAR (Ours)	**79.52 ± 2.80**	**73.98 ± 2.27**	**80.53 ± 1.21**	**90.80 ± 0.31**	**52.44 ± 2.60**	**85.83 ± 1.59**	**64.44 ± 1.44**	**80.28 ± 1.42**	**37.27 ± 8.78**	40.25 ± 0.77	**68.53 ± 2.32**

"–" indicates excessive runtime (mean execution time over 10 runs $>$ 7200 s).

TABLE VI

ABLATION STUDY RESULTS IN TERMS OF ACC, KAPPA, AND NMI (%). THE BEST RESULTS ARE HIGHLIGHTED IN **BOLD**.

Metric	Method	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	Avg.
ACC	w/o Granulation	74.06 ± 8.54	95.40 ± 0.41	92.66 ± 12.71	97.26 ± 0.10	69.70 ± 9.48	90.77 ± 2.65	70.52 ± 2.54	84.05 ± 1.22	90.98 ± 0.27	28.37 ± 1.92	79.38 ± 3.98
	w/o $\|\mathbf{Y}-\mathbf{YP}^t\|_F^2$	71.49 ± 8.67	78.82 ± 14.59	64.32 ± 5.57	96.27 ± 1.65	44.88 ± 4.39	88.18 ± 2.70	50.02 ± 4.37	75.52 ± 3.79	61.73 ± 6.71	26.28 ± 2.16	65.75 ± 5.46
	w/o Both	73.17 ± 9.02	80.77 ± 12.65	63.52 ± 5.27	96.21 ± 1.03	46.13 ± 4.23	88.55 ± 3.85	52.35 ± 2.72	75.22 ± 2.34	58.56 ± 4.95	26.74 ± 1.68	66.12 ± 4.77
	TEAR	**79.90 ± 1.87**	**95.59 ± 0.42**	**96.74 ± 0.37**	**97.32 ± 0.11**	**72.27 ± 3.34**	**91.52 ± 2.14**	**72.24 ± 4.05**	**85.38 ± 1.14**	**91.60 ± 1.90**	**29.11 ± 0.77**	**81.17 ± 1.61**
Kappa	w/o Granulation	68.01 ± 9.55	90.03 ± 0.90	84.51 ± 26.21	96.35 ± 0.13	53.95 ± 16.54	89.75 ± 2.94	64.50 ± 3.24	82.27 ± 1.35	58.27 ± 0.85	25.51 ± 1.99	71.31 ± 6.37
	w/o $\|\mathbf{Y}-\mathbf{YP}^t\|_F^2$	65.23 ± 9.60	59.25 ± 27.11	36.70 ± 8.18	95.03 ± 2.20	23.30 ± 4.49	86.86 ± 3.00	40.58 ± 5.55	72.79 ± 4.22	-9.19 ± 8.10	23.33 ± 2.24	49.39 ± 7.47
	w/o Both	67.00 ± 10.21	63.41 ± 22.46	35.44 ± 7.68	94.95 ± 1.37	30.23 ± 4.73	87.27 ± 4.28	43.14 ± 3.33	72.46 ± 2.60	-6.89 ± 10.50	23.81 ± 1.74	51.08 ± 6.89
	TEAR	**74.65 ± 2.34**	**90.44 ± 0.90**	**92.92 ± 0.78**	**96.42 ± 0.14**	**59.02 ± 4.87**	**90.58 ± 2.37**	**65.94 ± 5.15**	**83.75 ± 1.26**	**60.36 ± 6.58**	**26.27 ± 0.80**	**74.04 ± 2.52**
NMI	w/o Granulation	76.66 ± 4.02	72.79 ± 2.05	72.41 ± 25.20	90.65 ± 0.27	48.54 ± 9.25	85.52 ± 1.57	62.10 ± 2.11	78.82 ± 1.47	34.52 ± 0.80	39.72 ± 1.44	66.17 ± 4.82
	w/o $\|\mathbf{Y}-\mathbf{YP}^t\|_F^2$	74.69 ± 4.37	39.65 ± 25.58	24.69 ± 5.62	89.19 ± 2.17	25.93 ± 4.01	82.30 ± 1.96	44.49 ± 3.90	70.73 ± 1.66	3.75 ± 2.01	36.77 ± 1.49	49.22 ± 5.28
	w/o Both	76.17 ± 4.38	42.69 ± 21.33	23.19 ± 5.41	88.68 ± 1.95	29.63 ± 6.65	82.63 ± 2.50	49.14 ± 2.36	70.25 ± 1.75	4.04 ± 2.44	36.76 ± 1.22	50.32 ± 5.00
	TEAR	**79.52 ± 2.80**	**73.98 ± 2.27**	**80.53 ± 1.21**	**90.80 ± 0.31**	**52.44 ± 2.60**	**85.83 ± 1.59**	**64.44 ± 1.44**	**80.28 ± 1.42**	**37.27 ± 8.78**	**40.25 ± 0.77**	**68.53 ± 2.32**

material. From these results, we make the following observations:

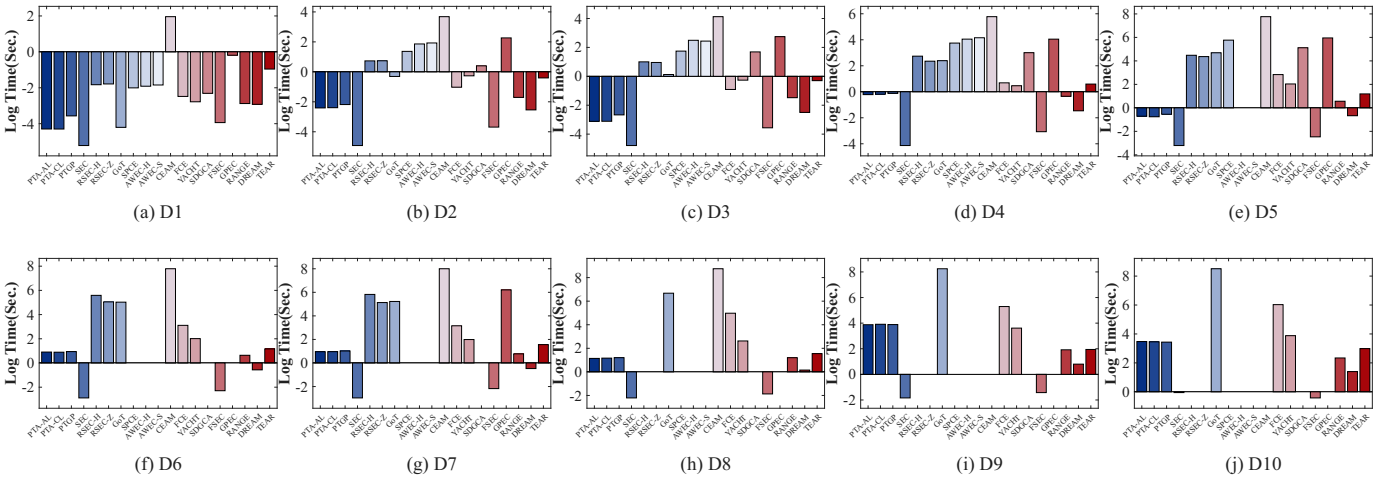
First, compared to the average performance of base clusterings, TEAR consistently achieves substantial gains in both ACC and Kappa across all 10 datasets, and improves NMI on 9 datasets. On Letter Recognition, although the NMI of TEAR is slightly lower than the base-clustering average, its standard deviation drops sharply from 8.29 to 0.77 percentage points, which is less than one-tenth of the original standard deviation, exhibiting substantially reduced fluctuation.

Second, when compared with the best base clustering within each ensemble, TEAR achieves higher scores on 8, 9, and 8 datasets in terms of ACC, Kappa, and NMI, respectively. For instance, on PD, TEAR elevates ACC from 73.10% to 85.38%, yielding a significant improvement of 12.28 percentage points. This indicates that TEAR successfully exploits complementary information across diverse base clusterings to generate a consensus partition that surpasses individual partition in the candidate pool.

Lastly, TEAR ranks first among all compared ensemble clustering methods on all 10 datasets in terms of ACC and Kappa. For NMI, it achieves the best performance on 9 datasets and ranks second on the remaining one. Among the methods that successfully execute across all 10 datasets, TEAR attains overall average scores of 81.17% ACC, 74.04% Kappa, and 68.53% NMI, outperforming PTGP by 5.43, 7.59, and 5.25 percentage points, respectively. TEAR also exhibits smaller standard deviations than most competing methods in most cases, indicating greater robustness across repeated runs. To further validate TEAR, a statistical significance analysis is provided in the supplementary material.

C. Ablation Study

To assess the contribution of each core module, we conduct ablation experiments on three variants: "w/o Granulation", "w/o $\|\mathbf{Y}-\mathbf{YP}^t\|_F^2$" (setting $\lambda = 0$), and "w/o Both" (without granulation and setting $\lambda = 0$). Table VI summarizes their results in terms of ACC, Kappa, and NMI. We observe that removing either module leads to varying degrees of performance degradation across all three metrics on all



(a) D1 (b) D2 (c) D3 (d) D4 (e) D5

(f) D6 (g) D7 (h) D8 (i) D9 (j) D10

Fig. 3. Log-transformed absolute runtime of compared methods and TEAR on the ten datasets.

datasets. Specifically, "w/o Granulation" reduces the average ACC, Kappa, and NMI from 81.17%, 74.04%, and 68.53% to 79.38%, 71.31%, and 66.17%, respectively. On Breast Cancer Wisconsin (D3), the decreases reach 4.08, 8.41, and 8.12 percentage points, highlighting the contribution of granulation. "w/o $\|\mathbf{Y}-\mathbf{Y}\mathbf{P}^t\|_F^2$" causes larger average decreases of 15.42, 24.65, and 19.31 percentage points in ACC, Kappa, and NMI, respectively. This effect is particularly pronounced on HTRU2 (D9), where ACC drops from 91.60% to 61.73% and Kappa drops from 60.36% to -9.19%. Moreover, TEAR exhibits smaller standard deviations than the three variants in most cases, indicating more robust clustering results.

Notably, "w/o Both" achieves higher average scores across all three metrics than "w/o $\|\mathbf{Y}-\mathbf{Y}\mathbf{P}^t\|_F^2$", indicating that granulation alone does not necessarily improve clustering performance. While suppressing NCAs, granulation may also sever some VWAs, introducing additional missing links. Meanwhile, TEAR outperforms "w/o Granulation", showing that granulation further improves clustering when combined with high-order self-expression. These results indicate that the two modules are complementary: granulation suppresses NCAs in the original base clusterings, while high-order self-expression helps recover missing links both inherent in the original base clusterings and introduced by granulation, preserving its benefits while mitigating connection loss. Further analysis of the effectiveness of granulation is provided in Section IV-E. Additional ablation results in terms of ARI, PUR, and F-score are provided in the supplemental material.

D. Runtime Comparison

Fig. 3 compares log-transformed runtimes on ten datasets. The times reported below are averages over ten runs in seconds. Missing bars indicate that the time limit was exceeded.

SEC and FSEC are relatively fast. For example, on Optdigits (D6), SEC and FSEC require 0.06 and 0.10 seconds, respectively, whereas TEAR requires 3.20 seconds. Although TEAR is somewhat slower, it substantially improves clustering performance, as shown in Section IV-B, making the additional runtime worthwhile.

In addition, TEAR exhibits broadly comparable runtime efficiency to other dynamic anchor-based methods across datasets. For example, on PD (D8), TEAR and YACHT require 4.71 and 13.79 seconds, respectively, while RANGE and DREAM require 3.33 and 1.17 seconds. Overall, TEAR is faster than YACHT but slower than RANGE and DREAM.

Furthermore, on larger datasets such as HTRU2 (D9) and Letter Recognition (D10), TEAR requires 6.99 and 19.81 seconds, respectively. In comparison, GoT requires 3821.51 and 4955.70 seconds, while FCE requires 199.65 and 414.41 seconds. Several methods fail to return results within the time limit. TEAR completes experiments on all ten datasets and achieves a favorable trade-off between runtime efficiency and clustering performance.

E. Granulation Analysis

To analyze the effect of adaptive granulation, we compare adaptive granulation (Adaptive) with random granulation (Random). Within each original base cluster, Random matches Adaptive in both granule count and granule-size distribution. We evaluate the VWA retention rate (VRR), NCA removal rate (NRR), and granulation ratio g/s. Using ground-truth labels, VRR measures the fraction of original VWAs retained after granulation, whereas NRR measures the fraction of original NCAs removed. Fig. 4 presents the results, with bars and error bars indicating means and standard deviations over ten runs.

Fig. 4(a) and (b) show that Adaptive generally achieves higher VRR and NRR than Random across the ten datasets. On Zoo (D1), Adaptive achieves 94.04% VRR and 57.33% NRR, compared with 81.30% and 40.59% for Random, respectively. These results demonstrate that ensemble-driven granulation more effectively suppresses NCAs while preserving VWAs. Nevertheless, VRR drops to approximately 75% on Sensor Readings 2 (D5) and Statlog (Landsat Satellite) (D7), highlighting the need for complementary high-order self-

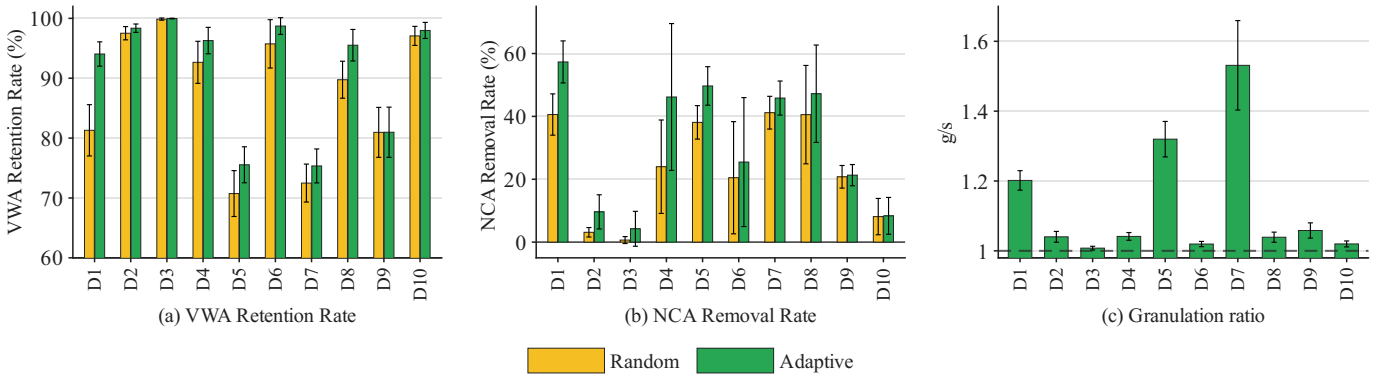


Fig. 4. Granulation analysis on the ten datasets.

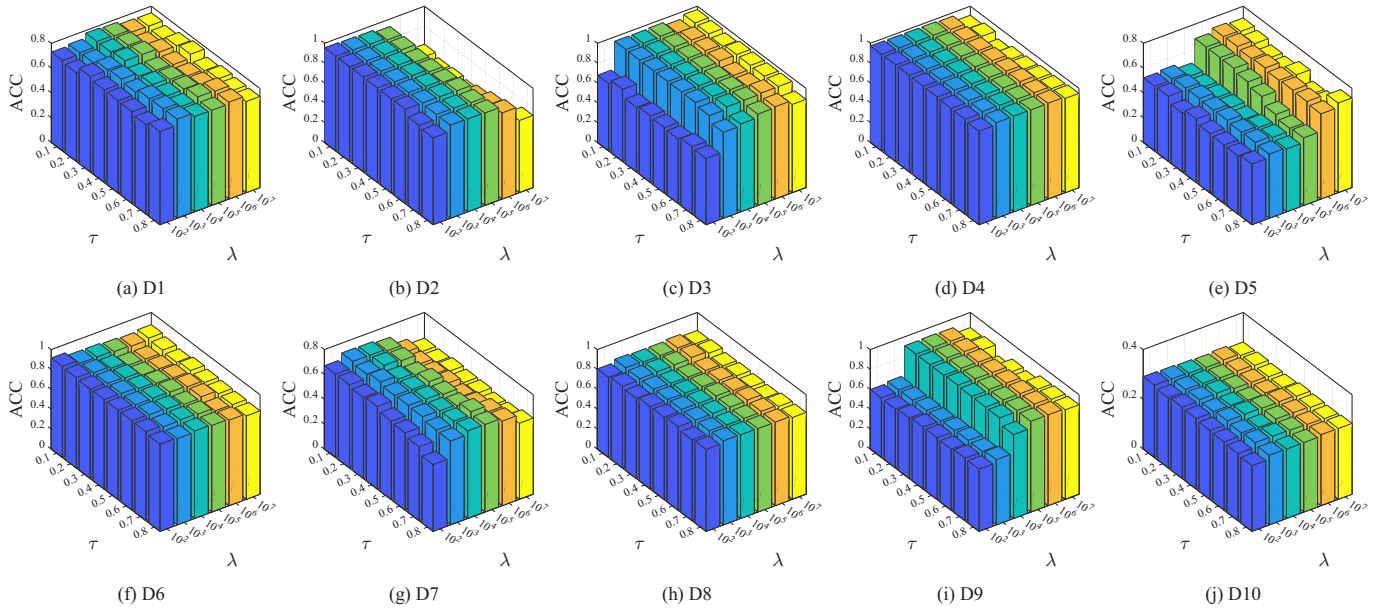


Fig. 5. Sensitivity of TEAR to τ and λ in terms of ACC on the ten datasets.

expression to exploit indirect structural relationships in global consensus learning.

Fig. 4(c) shows that the granulation ratio g/s ranges from approximately 1.01 to 1.53, remaining close to 1 on most datasets. This indicates that adaptive granulation selectively refines base clusters without substantially increasing the number of structural units in most cases. Overall, these results demonstrate that adaptive granulation effectively suppresses NCAs while largely preserving VWAs, with only a moderate increase in the number of granules.

F. Parameter Study

TEAR involves two hyperparameters: λ controls the strength of high-order self-expression, while τ controls granulation acceptance, with larger values encouraging more aggressive granulation. Fig. 5 shows their effects on ACC for $\lambda \in \{10^2, 10^3, \ldots, 10^7\}$ and $\tau \in \{0.1, 0.2, \ldots, 0.8\}$. TEAR maintains similar performance over a broad parameter region on most datasets, exhibiting good robustness to hyperparameter selection, particularly on Wireless Indoor Localization (D4) and PD (D8).

For λ, appropriately increasing its value improves performance on Breast Cancer Wisconsin (D3), Sensor_Readings_2 (D5), and HTRU2 (D9), demonstrating the benefit of high-order structural information. Conversely, Breast (Diagnostic) (D2) performs better with smaller values, indicating the need to balance high-order self-expression and anchor reconstruction.

For τ, performance generally remains stable under suitable values of λ, although increasing τ to 0.8 degrades performance under some parameter combinations on Breast Cancer Wisconsin (D3). This agrees with the intuition that moderate granulation can suppress NCAs, whereas excessive granulation may disrupt VWAs and introduce additional missing links. These observations highlight the need to balance structural refinement through granulation with the recovery of missing

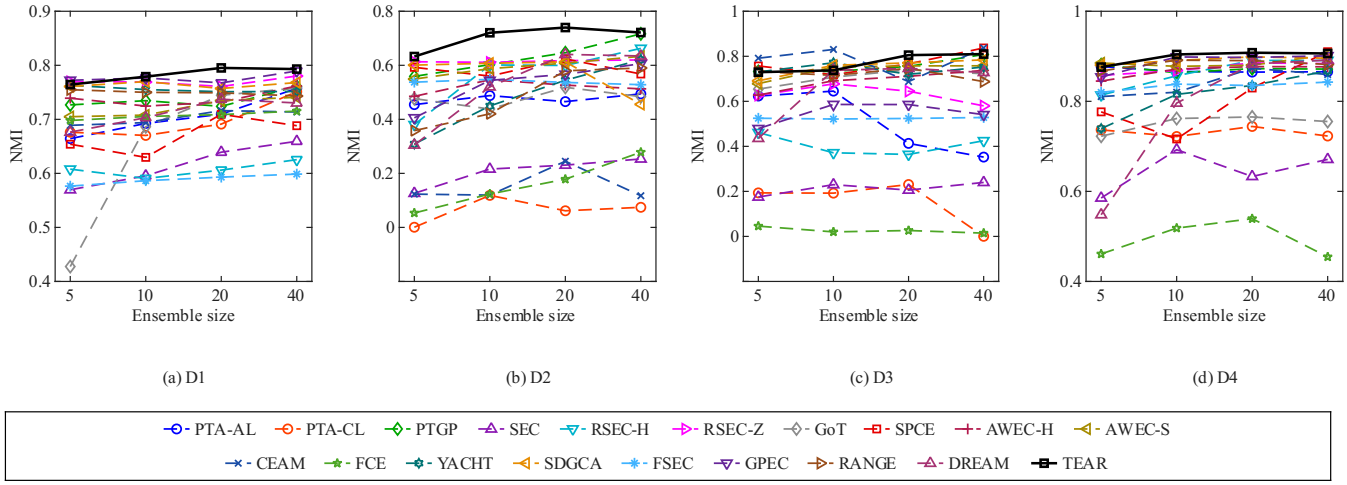


Fig. 6. Clustering performances of different algorithms by varying ensemble size w.r.t. NMI.

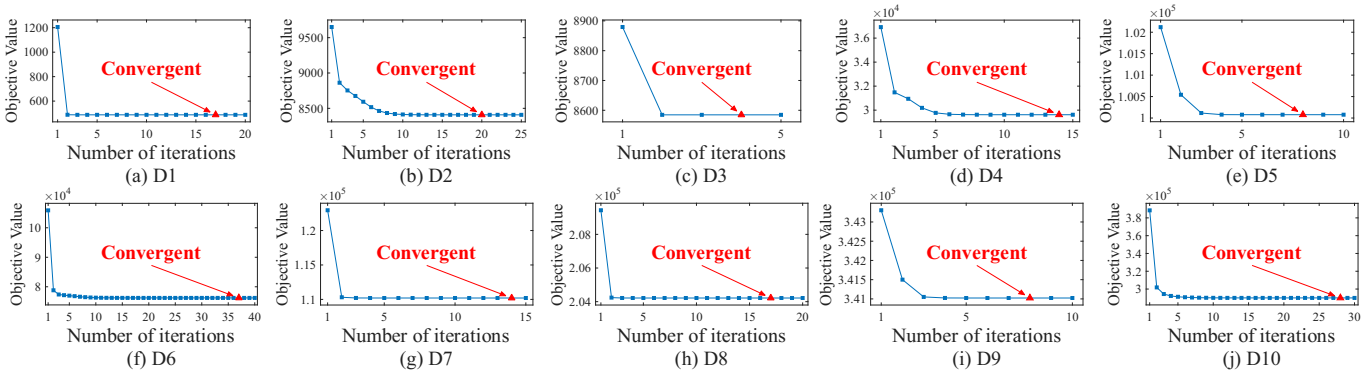


Fig. 7. Convergent curves of TEAR on datasets D1-D10.

links through high-order self-expression.

G. Influence of Ensemble Size

Fig. 6 compares NMI under different ensemble sizes on four datasets. TEAR generally benefits from larger ensembles, with performance improving from 5 to 10 or 20 base clusterings before stabilizing on D1, D2, and D4. On D3, the improvement is more pronounced between 10 and 20 base clusterings. However, performance does not always increase monotonically, as additional base clusterings may not improve ensemble quality or diversity.

H. Convergence Study

Fig. 7 shows that TEAR's objective decreases rapidly before stabilizing on all ten datasets, consistent with the convergence analysis in Section III-F. The curves exhibit a smooth downward trend, with most of the reduction occurring during the early iterations. All runs meet the stopping criterion within 40 iterations, demonstrating efficient convergence in practice.

V. Conclusion and Discussion

We propose TEAR, a linear-time ensemble clustering method that addresses the structural granularity bottleneck of existing dynamic anchor-based methods. Without accessing raw data features, TEAR constructs consensus pseudo-features and adaptively refines base clusters into granules using co-clustering support from other base clusterings, aiming to suppress NCAs while preserving VWAs. These granules provide a finer-grained structural basis for global consensus learning. TEAR then integrates granulated hypergraph reconstruction and high-order self-expression regularization into dynamic anchor learning to exploit indirect structural relationships and learn a global consensus representation. The final consensus partition is then obtained through spectral embedding and k-means on the learned representation. Cluster-level counting and low-rank computation enable linear-time complexity with respect to the number of samples. The alternating optimization further guarantees a non–increasing sequence of objective values. Experiments on 10 datasets against 18 representative SOTA methods demonstrate superior overall clustering performance and competitive scalability. Ablation studies and granulation analysis further demonstrate the effectiveness of adaptive granulation and its complementarity with high-order self-expression. Future work will explore jointly optimizing granulation and dynamic anchors in an end-to-end framework.

Generative AI Use Statement

Generative AI tools, including ChatGPT, were used for code debugging during experiment and language polishing, grammar checking, formatting assistance during manuscript preparation.

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Overview of the Supplemental Material

- **Accelerating Granulation:** derivation of the cluster-level counting formula, cluster-level counting pseudocode, and complexity analysis.
- **Additional Experiments:** supplementary results, including dataset and compared method details, additional clustering performance, statistical significance analysis, and ablation results.

SI. Accelerating Granulation

A. Equivalent Counting Formulation

For a candidate granulation $\{\mathcal{G}_1, \mathcal{G}_2\}$ of $\mathcal{G}$ in the p-th base clustering, we derive the counting form in Eq. (14) from the pairwise criterion in Eq. (2). Recall that $\rho_{a,b}^{(p)}$ is the proportion of the other base clusterings that assign samples x_a and x_b to the same cluster.

For each base clustering π^j with $j \neq p$, we construct two frequency vectors $\mathbf{q}_1$ and $\mathbf{q}_2$ with matching cluster indices, where

$$q_{h,r} = |\mathcal{G}_h \cap \mathcal{C}_r^j|, \qquad h = 1, 2.$$

The product $q_{1,r}q_{2,r}$ counts the sample pairs crossing the two sub-granules that belong to cluster $\mathcal{C}_r^j$. Summing these products over clusters gives

$$b_j = \sum_r q_{1,r}q_{2,r} = \mathbf{q}_1^\top \mathbf{q}_2.$$

The two frequency vectors are recomputed for each π^j; their dependence on j is omitted for simplicity. Expanding the support rate in Eq. (2) and exchanging the summations replaces pairwise enumeration with the counts b_j, giving

$$\begin{aligned} \Delta &= 2 \sum_{a:\, x_a \in \mathcal{G}_1} \sum_{b:\, x_b \in \mathcal{G}_2} \rho_{a,b}^{(p)} - \tau m \\ &= \frac{2}{v-1} \sum_{j \neq p} b_j - \tau m, \end{aligned} \tag{S1}$$

where $m = |\mathcal{G}_1||\mathcal{G}_2|$. Therefore, cluster-level counting evaluates exactly the same cost change as pairwise enumeration.

B. Cluster-Level Counting Algorithm

Algorithm S1 evaluates one candidate granulation. For each base clustering π^j with $j \neq p$, it constructs two sparse frequency vectors $\mathbf{q}_1$ and $\mathbf{q}_2$, implemented as hash tables storing only nonzero counts. Entries absent from a hash table have count zero. Candidate generation and recursive granulation follow the procedure in the main text.

For each $j \neq p$, constructing the two frequency vectors visits $|\mathcal{G}|$ samples, and accumulating b_j visits at most $|\mathcal{G}_1|$ nonzero entries in $\mathbf{q}_1$. With expected constant-time hash-table access, the expected scoring cost is $\mathcal{O}(v|\mathcal{G}|)$, without enumerating the m sample pairs. Processing the other base clusterings sequentially requires $\mathcal{O}(|\mathcal{G}|)$ auxiliary space, excluding the input ensemble. Only nonzero frequencies are stored; no dense vector over all base-cluster labels is required.

Algorithm S1 Cluster-Level Counting

Require: Base clustering results Π, index p, sub-granules $\mathcal{G}_1, \mathcal{G}_2$, threshold τ
Ensure: Cost change Δ

$m \leftarrow |\mathcal{G}_1||\mathcal{G}_2|$, $b \leftarrow 0$
for $j \in \{1, \ldots, v\} \setminus \{p\}$ **do**
  Initialize sparse frequency vectors $\mathbf{q}_1, \mathbf{q}_2$ as empty hash tables with zero default counts
  for $h = 1, 2$ **do**
    for each $x_a \in \mathcal{G}_h$ **do**
      Set r to the cluster index of x_a in π^j
      Increment $q_{h,r}$ by one
    end for
  end for
  $b_j \leftarrow 0$
  for each nonzero entry $q_{1,r}$ in $\mathbf{q}_1$ **do**
    $b_j \leftarrow b_j + q_{1,r}q_{2,r}$
  end for
  $b \leftarrow b + b_j$
end for
$\Delta \leftarrow 2b/(v-1) - \tau m$ by Eq. (S1)
return Δ

SII. Additional Experiments

All experiments reported in the main paper and this supplemental material were conducted in MATLAB R2024a on a shared Windows server equipped with two Intel Xeon Gold 6248R CPUs at 3.00 GHz and 768 GB RAM. The random seed was fixed using `rng(2026)`.

A. Details of Datasets

The ten datasets used in our experiments cover biology, healthcare, sensor measurements, remote sensing, astronomy, and character recognition. Their statistics and sources are summarized in Table S1. We briefly describe each dataset below.

- **Zoo**[2]: This dataset describes animals using attributes such as hair, feathers, habitat, and number of legs. The samples belong to seven animal categories.
- **Breast (Diagnostic)**[3]: This dataset contains features computed from digitized images of breast mass fine-needle aspirates. The features characterize cell nuclei, and the labels indicate benign or malignant diagnoses.
- **Breast Cancer Wisconsin**[4]: This dataset contains cell-level attributes, including cell size, shape, and nuclear characteristics, for benign and malignant breast tumor samples.
- **Wireless Indoor Localization**[5]: This dataset contains signal strengths from seven Wi-Fi sources measured on a smartphone. The labels correspond to four indoor rooms.

[2] https://archive.ics.uci.edu/dataset/111/zoo
[3] https://archive.ics.uci.edu/dataset/17/breast+cancer+wisconsin+diagnostic
[4] https://archive.ics.uci.edu/dataset/15/breast+cancer+wisconsin+original
[5] https://archive.ics.uci.edu/dataset/422/wireless+indoor+localization

TABLE S1
DETAILS OF THE DATASETS USED IN THE EXPERIMENTS.

No.	Dataset	#Sample	#Feature	#Class	Scale	Domain
D1	Zoo	101	16	7	Small	Biology
D2	Breast (Diagnostic)	569	30	2	Small	Healthcare
D3	Breast Cancer Wisconsin	699	9	2	Small	Healthcare
D4	Wireless Indoor Localization	2,000	7	4	Medium	Sensor
D5	Sensor_Readings_2	5,456	2	4	Medium	Sensor
D6	Optdigits	5,620	64	10	Medium	Character recognition
D7	Statlog (Landsat Satellite)	6,435	36	6	Medium	Remote sensing
D8	PD	10,992	16	10	Large	Character recognition
D9	HTRU2	17,898	8	2	Large	Astronomy
D10	Letter Recognition	20,000	16	26	Large	Character recognition

- **Sensor_Readings_2**[6]: This dataset contains measurements collected by a mobile robot following a wall. Its two features describe the simplified distances in front of and to the left of the robot, and its labels indicate four movement directions.
- **Optdigits**[7]: This dataset contains optical images of handwritten digits represented by 64 features computed from pixel counts in an 8×8 grid. The labels correspond to digits from 0 to 9.
- **Statlog (Landsat Satellite)**[8]: This remote sensing dataset contains multispectral values from 3×3 pixel neighborhoods in satellite images. Its labels describe the land-cover type of the central pixel.
- **PD**[9]: The Pen-Based Recognition of Handwritten Digits dataset represents handwritten digits using 16 features derived from pen coordinates. Its ten classes correspond to digits from 0 to 9.
- **HTRU2**[10]: This dataset contains pulsar candidates collected during the High Time Resolution Universe survey. Eight features describe the candidate signals, which are labeled as pulsar or non-pulsar candidates.
- **Letter Recognition**[11]: This dataset contains features of character images generated from different fonts with random distortions. Its 26 classes correspond to the uppercase English letters.

B. Compared Methods

We compare TEAR with the following ensemble clustering methods.

- **PTA-AL, PTA-CL, and PTGP** [19]: These methods use probability trajectories on a sparse graph to measure similarities between micro-clusters. PTA-AL and PTA-CL apply average-link and complete-link hierarchical clustering, respectively, whereas PTGP partitions a weighted micro-cluster–cluster bipartite graph.
- **SEC** [30]: Spectral ensemble clustering exploits the equivalence between spectral clustering on the co-association matrix and weighted k-means to obtain the consensus partition efficiently.
- **RSEC-H and RSEC-Z** [31]: RSEC learns a low-rank representation of the co-association matrix while performing spectral clustering. RSEC-H applies k-means to the learned partition matrix, whereas RSEC-Z applies spectral clustering to the learned representation.
- **GoT** [8]: GoT refines the co-association matrix using shortest paths, identifies representative prototypes, and constructs growing trees to obtain the consensus partition.
- **SPCE** [32]: Self-paced clustering ensemble gradually incorporates samples from easy to difficult into consensus learning to reduce the influence of unreliable instances.
- **AWEC-H and AWEC-S** [22]: AWEC learns an enhanced connection matrix using adaptively weighted high-order connectivity. AWEC-H and AWEC-S obtain the consensus through hierarchical and spectral clustering, respectively.
- **CEAM** [33]: CEAM represents base clusterings as an adaptive multiplex and diffuses their representations using manifold ranking to obtain a consensus representation.
- **FCE** [34]: Fair clustering ensemble incorporates an equal cluster capacity constraint into consensus learning to address group-level fairness.
- **YACHT** [14]: YACHT represents base clusterings as a hypergraph and jointly learns dynamic anchors and sample representations through hypergraph reconstruction.
- **SDGCA** [18]: SDGCA uses similarity and dissimilarity information to construct an enhanced co-association matrix for consensus clustering.
- **FSEC** [13]: FSEC uses an anchor graph and singular value decomposition to accelerate spectral clustering for ensemble construction and consensus learning.
- **GPEC** [35]: GPEC is derived from a generalization analysis of ensemble clustering and assigns weights to base clusterings to account for their bias and diversity.
- **RANGE** [16]: RANGE integrates outlier decoupling into

[6]https://archive.ics.uci.edu/dataset/194/wall+following+robot+navigation+data

[7]https://archive.ics.uci.edu/dataset/80/optdigits

[8]https://archive.ics.uci.edu/dataset/146/statlog+(landsat+satellite)

[9]https://archive.ics.uci.edu/dataset/81/pen+based+recognition+of+handwritten+digits

[10]https://archive.ics.uci.edu/dataset/372/htru2

[11]https://archive.ics.uci.edu/dataset/59/letter+recognition

TABLE S2
THE CLUSTERING PERFORMANCES OF DIFFERENT ALGORITHMS ARE MEASURED BY ARI(%). BEST RESULTS ARE HIGHLIGHTED IN **BOLD**, AND SECOND-BEST RESULTS ARE MARKED WITH AN UNDERLINE.

Method	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	Avg.
k-means (best)	83.90 ± 0.00	73.02 ± 0.00	82.84 ± 0.00	85.34 ± 0.00	33.95 ± 0.00	75.76 ± 0.00	47.88 ± 0.00	64.05 ± 0.00	53.16 ± 0.00	16.47 ± 0.00	61.64 ± 0.00
k-means (average)	52.88 ± 18.65	21.69 ± 14.50	39.63 ± 18.47	32.74 ± 19.59	17.01 ± 8.34	40.93 ± 15.43	23.54 ± 10.12	34.65 ± 13.21	2.02 ± 4.51	14.43 ± 2.30	27.95 ± 12.51
PTA-AL [19]	56.97 ± 10.22	49.53 ± 17.04	31.61 ± 37.83	84.82 ± 8.77	27.83 ± 21.57	78.90 ± 5.42	52.84 ± 2.48	64.56 ± 4.27	50.95 ± 8.99	13.72 ± 3.12	51.17 ± 11.97
PTA-CL [19]	52.86 ± 11.49	3.30 ± 7.72	6.65 ± 8.78	61.10 ± 14.48	7.69 ± 5.46	35.52 ± 12.71	-0.00 ± 0.07	33.20 ± 9.99	-0.01 ± 0.00	0.04 ± 0.05	20.03 ± 7.07
PTGP [19]	58.11 ± 9.98	72.07 ± 14.91	84.86 ± 2.99	86.30 ± 6.43	33.99 ± 14.27	81.95 ± 3.71	48.89 ± 4.12	64.81 ± 5.17	46.37 ± 1.30	13.34 ± 3.80	59.07 ± 6.67
SEC [30]	45.89 ± 15.64	17.65 ± 18.79	21.04 ± 23.95	47.69 ± 17.55	14.03 ± 12.02	37.87 ± 10.32	22.61 ± 13.64	23.57 ± 9.33	37.25 ± 29.10	6.70 ± 2.61	27.43 ± 15.30
RSEC-Z [31]	65.30 ± 8.89	72.50 ± 2.60	73.06 ± 19.76	89.71 ± 1.64	25.67 ± 6.35	78.84 ± 3.41	43.12 ± 7.91	–	–	–	–
RSEC-H [31]	44.13 ± 15.91	67.12 ± 16.95	26.32 ± 33.12	90.85 ± 1.71	26.87 ± 8.62	75.41 ± 3.88	36.20 ± 7.88	–	–	–	–
GoT [8]	63.38 ± 6.04	57.81 ± 19.75	83.78 ± 3.90	65.23 ± 11.30	14.69 ± 14.51	62.08 ± 10.39	53.50 ± 0.79	52.08 ± 6.77	46.76 ± 1.68	12.13 ± 1.42	51.15 ± 7.65
SPCE [32]	63.25 ± 8.93	72.85 ± 2.85	86.12 ± 2.01	80.11 ± 13.59	25.88 ± 12.82	–	–	–	–	–	–
AWEC-H [22]	63.42 ± 4.58	60.98 ± 7.90	75.85 ± 20.80	88.37 ± 3.17	–	–	–	–	–	–	–
AWEC-S [22]	64.21 ± 6.38	71.70 ± 3.91	81.37 ± 21.56	87.80 ± 7.11	–	–	–	–	–	–	–
CEAM [33]	54.04 ± 6.53	29.98 ± 29.42	78.96 ± 6.65	88.12 ± 4.74	34.94 ± 6.36	63.31 ± 7.96	31.27 ± 5.52	47.33 ± 4.55	–	–	–
FCE [34]	51.52 ± 3.52	20.16 ± 17.08	3.00 ± 1.67	49.17 ± 13.21	11.57 ± 4.08	76.68 ± 7.65	20.04 ± 3.90	50.13 ± 3.82	-0.00 ± 0.00	13.38 ± 1.26	29.56 ± 5.62
YACHT [14]	67.22 ± 1.83	66.15 ± 7.05	80.15 ± 19.95	83.14 ± 10.26	30.85 ± 5.13	73.61 ± 6.73	47.53 ± 6.28	57.39 ± 4.99	-4.45 ± 2.34	**16.54 ± 0.86**	51.81 ± 6.54
SDGCA [18]	67.05 ± 4.24	70.24 ± 11.07	85.61 ± 3.29	91.59 ± 2.90	14.41 ± 14.61	–	–	–	–	–	–
FSEC [13]	35.53 ± 14.55	59.39 ± 17.55	55.64 ± 13.62	80.82 ± 11.24	34.87 ± 6.16	75.72 ± 9.32	54.76 ± 6.49	67.98 ± 7.15	39.27 ± 4.81	14.79 ± 1.95	51.88 ± 9.28
GPEC [35]	67.39 ± 2.56	66.45 ± 11.36	57.39 ± 39.37	91.65 ± 1.30	34.76 ± 4.84	–	40.39 ± 6.22	–	–	–	–
RANGE [16]	61.20 ± 8.28	68.66 ± 26.73	80.43 ± 30.63	90.50 ± 13.74	30.97 ± 6.12	80.23 ± 2.95	49.11 ± 6.44	67.61 ± 3.60	2.99 ± 1.39	16.13 ± 1.23	54.78 ± 10.11
DREAM [15]	68.26 ± 12.06	75.21 ± 3.57	84.67 ± 2.00	90.45 ± 7.97	31.25 ± 5.60	78.46 ± 4.30	49.75 ± 6.09	61.27 ± 2.77	-6.13 ± 2.13	15.36 ± 0.84	54.85 ± 4.73
TEAR (Ours)	**70.72 ± 1.91**	**83.01 ± 1.53**	**87.29 ± 1.39**	**93.07 ± 0.26**	**49.47 ± 1.74**	**83.43 ± 3.06**	**57.37 ± 3.82**	**72.44 ± 1.93**	**54.06 ± 7.81**	15.89 ± 0.69	**66.68 ± 2.41**

"–" indicates excessive runtime (mean execution time over 10 runs $>$ 7200 s).

TABLE S3
THE CLUSTERING PERFORMANCES OF DIFFERENT ALGORITHMS ARE MEASURED BY PUR(%). BEST RESULTS ARE HIGHLIGHTED IN **BOLD**, AND SECOND-BEST RESULTS ARE MARKED WITH AN UNDERLINE.

Method	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	Avg.
k-means (best)	93.07 ± 0.00	96.31 ± 0.00	97.57 ± 0.00	97.60 ± 0.00	93.35 ± 0.00	96.03 ± 0.00	87.55 ± 0.00	96.38 ± 0.00	97.57 ± 0.00	57.84 ± 0.00	91.33 ± 0.00
k-means (average)	75.37 ± 13.53	93.06 ± 1.91	96.72 ± 0.43	95.58 ± 5.25	87.56 ± 7.31	89.23 ± 13.88	82.65 ± 9.03	85.12 ± 16.54	97.28 ± 0.48	43.56 ± 11.64	84.61 ± 8.00
PTA-AL [19]	80.99 ± 2.13	85.15 ± 5.59	76.41 ± 14.06	91.79 ± 7.75	64.04 ± 14.30	87.66 ± 4.62	73.61 ± 2.36	76.37 ± 4.56	91.86 ± 1.32	29.43 ± 1.86	75.73 ± 5.85
PTA-CL [19]	79.50 ± 2.76	64.78 ± 4.25	67.27 ± 3.68	71.71 ± 8.89	50.27 ± 7.82	54.62 ± 12.44	24.05 ± 0.28	51.55 ± 6.62	90.84 ± 0.00	4.56 ± 0.42	55.92 ± 4.72
PTGP [19]	81.78 ± 2.56	92.34 ± 4.60	96.09 ± 0.80	93.41 ± 5.76	67.42 ± 9.81	90.32 ± 3.13	71.56 ± 3.18	77.21 ± 5.09	90.84 ± 0.00	28.12 ± 3.09	78.91 ± 3.80
SEC [30]	75.54 ± 6.57	71.44 ± 8.82	74.54 ± 8.52	68.17 ± 11.85	58.19 ± 7.97	62.09 ± 5.66	54.00 ± 10.50	51.76 ± 8.34	**93.65 ± 2.07**	24.58 ± 2.07	63.40 ± 7.24
RSEC-Z [31]	84.75 ± 2.25	92.64 ± 0.75	92.25 ± 7.44	95.95 ± 0.67	67.14 ± 2.34	89.08 ± 2.59	68.82 ± 5.21	–	–	–	–
RSEC-H [31]	74.65 ± 4.90	90.65 ± 5.81	73.92 ± 12.38	96.41 ± 0.70	68.62 ± 5.88	86.56 ± 3.40	64.56 ± 5.03	–	–	–	–
GoT [8]	82.67 ± 2.35	87.77 ± 6.32	95.81 ± 1.04	77.46 ± 7.91	53.94 ± 9.91	74.27 ± 8.38	74.71 ± 0.39	68.08 ± 5.45	90.85 ± 0.03	30.00 ± 1.33	73.55 ± 4.31
SPCE [32]	80.69 ± 5.91	92.74 ± 0.83	96.44 ± 0.53	88.67 ± 10.84	60.84 ± 10.03	–	–	–	–	–	–
AWEC-H [22]	82.38 ± 1.30	89.03 ± 2.60	93.05 ± 7.18	95.28 ± 1.40	–	–	–	–	–	–	–
AWEC-S [22]	82.18 ± 3.55	92.37 ± 1.13	94.55 ± 7.69	94.07 ± 6.00	–	–	–	–	–	–	–
CEAM [33]	81.98 ± 2.28	75.31 ± 12.68	94.45 ± 1.84	95.17 ± 2.07	67.97 ± 3.67	75.98 ± 5.63	56.81 ± 5.40	64.90 ± 3.71	–	–	–
FCE [34]	81.39 ± 3.12	71.60 ± 8.50	65.52 ± 0.00	72.17 ± 9.79	56.76 ± 4.40	86.70 ± 5.92	50.66 ± 4.86	68.87 ± 3.51	90.84 ± 0.00	26.42 ± 1.43	67.09 ± 4.15
YACHT [14]	83.17 ± 2.14	90.67 ± 2.22	94.31 ± 7.01	92.30 ± 5.78	72.39 ± 3.79	84.98 ± 5.45	71.00 ± 3.94	73.51 ± 3.23	90.84 ± 0.00	30.00 ± 1.13	78.32 ± 3.47
SDGCA [18]	83.86 ± 1.75	91.88 ± 3.42	96.29 ± 0.88	96.67 ± 1.27	48.83 ± 9.99	–	–	–	–	–	–
FSEC [13]	70.79 ± 8.05	88.40 ± 5.57	87.07 ± 4.68	90.16 ± 8.42	73.65 ± 2.55	86.36 ± 6.14	75.28 ± 3.83	80.82 ± 5.39	90.84 ± 0.00	31.44 ± 1.04	77.48 ± 4.57
GPEC [35]	84.55 ± 1.88	90.63 ± 3.94	85.85 ± 14.38	96.74 ± 0.53	73.92 ± 2.65	–	65.68 ± 3.24	–	–	–	–
RANGE [16]	84.06 ± 3.02	91.44 ± 10.95	94.16 ± 11.36	96.22 ± 8.71	72.69 ± 5.14	89.47 ± 3.11	73.08 ± 4.08	81.61 ± 3.02	90.84 ± 0.00	29.98 ± 1.38	80.36 ± 5.08
DREAM [15]	79.60 ± 4.53	93.41 ± 1.05	96.05 ± 0.53	96.25 ± 3.48	72.80 ± 4.12	88.68 ± 3.88	72.47 ± 3.26	77.89 ± 2.28	90.84 ± 0.00	29.76 ± 0.78	79.78 ± 2.39
TEAR (Ours)	**86.63 ± 2.61**	**95.59 ± 0.42**	**96.74 ± 0.37**	**97.32 ± 0.11**	**76.20 ± 0.87**	**91.52 ± 2.14**	**78.24 ± 2.39**	**85.38 ± 1.14**	91.64 ± 1.88	**31.81 ± 0.80**	**83.11 ± 1.27**

"–" indicates excessive runtime (mean execution time over 10 runs $>$ 7200 s).

scalable anchor-based ensemble clustering to reduce the influence of outliers on consensus learning.

- **DREAM** [15]: DREAM combines dynamic anchor-based hypergraph reconstruction and one-step learning to obtain discrete consensus labels.

C. Clustering Performance Comparison

In the supplemental material, we further report clustering performance in terms of Adjusted Rand Index (ARI), Purity (PUR), and F-score, as shown in Tables S2, S3, and S4, respectively. TEAR achieves the best ARI on nine datasets and ranks third on D10 with an ARI of 15.89%. It also achieves the best PUR on nine datasets and ranks third on D9 with a PUR of 91.64%. These results demonstrate consistent clustering quality in terms of pairwise agreement and cluster homogeneity. Regarding F-score, TEAR achieves an average of 76.07% across the ten datasets, outperforming PTGP (71.20%) and DREAM (69.57%) by 4.87 and 6.50 percentage points, respectively. Overall, TEAR demonstrates superior clustering performance across multiple evaluation metrics, further validating its effectiveness in generating high-

TABLE S4

THE CLUSTERING PERFORMANCES OF DIFFERENT ALGORITHMS ARE MEASURED BY F-SCORE(%). BEST RESULTS ARE HIGHLIGHTED IN **BOLD**, AND SECOND-BEST RESULTS ARE MARKED WITH AN UNDERLINE.

Method	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	Avg.
k-means (best)	87.49 ± 0.00	87.68 ± 0.00	92.31 ± 0.00	89.02 ± 0.00	58.03 ± 0.00	78.20 ± 0.00	55.51 ± 0.00	67.39 ± 0.00	90.67 ± 0.00	18.56 ± 0.00	72.49 ± 0.00
k-means (average)	65.08 ± 11.64	37.74 ± 17.37	57.28 ± 16.48	38.79 ± 20.35	26.14 ± 13.39	44.09 ± 15.35	29.24 ± 12.19	38.20 ± 13.62	10.22 ± 12.44	16.67 ± 1.49	36.35 ± 13.43
PTA-AL [19]	66.01 ± 8.52	78.87 ± 6.68	70.19 ± 16.08	88.86 ± 6.10	55.84 ± 10.33	81.11 ± 4.75	61.72 ± 1.97	68.63 ± 3.60	89.30 ± 3.29	18.08 ± 2.60	67.86 ± 6.39
PTA-CL [19]	62.43 ± 9.56	68.07 ± 3.18	61.00 ± 3.79	73.18 ± 8.90	51.46 ± 1.25	45.28 ± 9.98	31.81 ± 0.15	43.26 ± 7.83	90.92 ± 0.00	7.43 ± 0.04	53.48 ± 4.47
PTGP [19]	66.77 ± 8.19	87.74 ± 5.67	93.13 ± 1.29	89.85 ± 4.48	58.51 ± 7.58	83.79 ± 3.29	58.29 ± 3.31	68.82 ± 4.44	87.43 ± 0.57	17.65 ± 3.17	71.20 ± 4.20
SEC [30]	58.45 ± 11.97	67.29 ± 7.47	73.94 ± 6.57	63.90 ± 10.51	50.29 ± 4.34	46.40 ± 8.31	41.76 ± 7.46	34.86 ± 7.08	**91.29 ± 5.87**	12.01 ± 2.02	54.02 ± 7.16
RSEC-Z [31]	72.27 ± 7.46	87.43 ± 1.19	87.90 ± 9.36	92.28 ± 1.23	49.21 ± 5.13	80.97 ± 3.06	53.71 ± 6.40	–	–	–	–
RSEC-H [31]	56.80 ± 11.39	84.62 ± 8.42	67.00 ± 14.66	93.14 ± 1.28	49.41 ± 6.63	77.95 ± 3.42	48.54 ± 6.17	–	–	–	–
GoT [8]	71.29 ± 5.02	81.78 ± 7.94	92.72 ± 1.67	75.23 ± 7.88	51.61 ± 6.00	66.34 ± 8.94	61.97 ± 0.64	57.66 ± 5.79	88.94 ± 0.40	16.79 ± 1.17	66.43 ± 4.54
SPCE [32]	72.51 ± 6.26	87.63 ± 1.24	93.72 ± 0.87	85.64 ± 9.32	56.47 ± 3.51	–	–	–	–	–	–
AWEC-H [22]	70.98 ± 3.48	81.64 ± 4.14	88.74 ± 9.76	91.29 ± 2.36	–	–	–	–	–	–	–
AWEC-S [22]	71.88 ± 4.71	86.73 ± 1.89	91.43 ± 9.99	90.98 ± 4.99	–	–	–	–	–	–	–
CEAM [33]	62.45 ± 5.41	66.90 ± 13.78	90.41 ± 2.99	91.10 ± 3.54	57.50 ± 5.23	67.18 ± 7.04	44.50 ± 4.74	53.24 ± 3.93	–	–	–
FCE [34]	60.34 ± 2.97	61.27 ± 8.29	53.66 ± 0.80	61.86 ± 9.91	36.96 ± 2.91	79.01 ± 6.88	34.21 ± 3.21	55.11 ± 3.44	62.51 ± 0.00	16.71 ± 1.21	52.16 ± 3.96
YACHT [14]	73.81 ± 1.48	84.09 ± 3.51	90.84 ± 9.55	87.44 ± 7.57	51.80 ± 4.32	76.32 ± 5.98	57.38 ± 5.01	61.94 ± 4.31	65.42 ± 2.86	**19.91 ± 0.82**	66.90 ± 4.54
SDGCA [18]	74.24 ± 3.23	86.69 ± 4.44	93.47 ± 1.39	93.70 ± 2.17	55.89 ± 6.01	–	–	–	–	–	–
FSEC [13]	48.63 ± 11.40	82.63 ± 6.40	78.96 ± 6.60	85.93 ± 7.87	54.57 ± 4.56	78.33 ± 8.09	63.37 ± 5.21	71.38 ± 6.27	84.39 ± 2.24	18.71 ± 1.75	66.69 ± 6.04
GPEC [35]	74.32 ± 1.86	84.12 ± 5.60	81.06 ± 17.21	93.74 ± 0.97	55.23 ± 3.06	–	51.59 ± 4.91	–	–	–	–
RANGE [16]	68.81 ± 7.09	85.20 ± 13.12	91.03 ± 14.02	92.88 ± 10.26	51.69 ± 3.90	82.23 ± 2.64	58.62 ± 5.29	70.95 ± 3.18	65.50 ± 0.63	19.53 ± 1.20	68.64 ± 6.13
DREAM [15]	75.73 ± 8.86	88.54 ± 1.71	93.09 ± 0.87	92.83 ± 5.85	51.97 ± 3.59	80.63 ± 3.84	59.06 ± 4.69	65.31 ± 2.39	69.61 ± 3.92	18.87 ± 0.80	69.57 ± 3.65
TEAR (Ours)	**76.75 ± 1.48**	**92.16 ± 0.71**	**94.14 ± 0.66**	**94.80 ± 0.20**	**66.99 ± 1.43**	**85.10 ± 2.73**	**65.62 ± 3.16**	**75.29 ± 1.73**	90.30 ± 2.22	19.53 ± 0.63	**76.07 ± 1.49**

"–" indicates excessive runtime (mean execution time over 10 runs $>$ 7200 s).

TABLE S5

ABLATION STUDY RESULTS IN TERMS OF ARI, PUR, AND F-SCORE (%). THE BEST RESULTS ARE HIGHLIGHTED IN **BOLD**.

Metric	Method	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	Avg.
ARI	w/o Granulation	64.33 ± 10.31	82.31 ± 1.51	78.50 ± 27.11	92.94 ± 0.25	43.87 ± 13.37	82.76 ± 3.24	53.13 ± 2.65	70.03 ± 2.17	51.57 ± 0.97	15.32 ± 1.47	63.48 ± 6.31
	w/o $\|\mathbf{Y}-\mathbf{YP}^t\|_F^2$	61.21 ± 10.07	40.51 ± 33.70	7.66 ± 8.17	90.69 ± 3.61	14.00 ± 4.40	77.92 ± 3.43	25.69 ± 7.90	60.03 ± 2.96	-4.57 ± 2.29	15.18 ± 1.73	38.83 ± 7.83
	w/o Both	62.62 ± 9.79	43.24 ± 31.27	6.57 ± 7.29	90.44 ± 2.39	20.36 ± 6.59	78.55 ± 4.71	31.69 ± 4.22	59.54 ± 2.85	-3.22 ± 2.36	15.32 ± 1.47	40.51 ± 7.30
	TEAR	**70.72 ± 1.91**	**83.01 ± 1.53**	**87.29 ± 1.39**	**93.07 ± 0.26**	**49.47 ± 1.74**	**83.43 ± 3.06**	**57.37 ± 3.82**	**72.44 ± 1.93**	**54.06 ± 7.81**	**15.89 ± 0.69**	**66.68 ± 2.41**
PUR	w/o Granulation	85.35 ± 2.13	95.40 ± 0.41	93.56 ± 9.86	97.26 ± 0.10	72.20 ± 9.96	90.77 ± 2.65	74.39 ± 1.50	84.05 ± 1.22	91.03 ± 0.19	31.12 ± 1.40	81.51 ± 2.94
	w/o $\|\mathbf{Y}-\mathbf{YP}^t\|_F^2$	84.06 ± 3.25	79.05 ± 14.30	67.53 ± 3.32	96.27 ± 1.65	61.37 ± 5.43	88.31 ± 2.47	58.92 ± 4.45	76.08 ± 3.01	90.84 ± 0.00	28.46 ± 1.76	73.09 ± 3.96
	w/o Both	85.54 ± 3.37	80.77 ± 12.65	67.10 ± 2.70	96.21 ± 1.03	65.67 ± 5.17	88.59 ± 3.75	61.17 ± 3.50	75.41 ± 2.19	90.84 ± 0.00	28.64 ± 1.36	74.00 ± 3.57
	TEAR	**86.63 ± 2.61**	**95.59 ± 0.42**	**96.74 ± 0.37**	**97.32 ± 0.11**	**76.20 ± 0.87**	**91.52 ± 2.14**	**78.24 ± 2.39**	**85.38 ± 1.14**	**91.64 ± 1.88**	**31.81 ± 0.80**	**83.11 ± 1.27**
F-score	w/o Granulation	71.37 ± 8.67	91.82 ± 0.69	89.96 ± 12.92	94.70 ± 0.19	64.61 ± 4.94	84.51 ± 2.89	61.74 ± 2.07	73.14 ± 1.92	89.58 ± 0.30	18.98 ± 1.34	74.04 ± 3.59
	w/o $\|\mathbf{Y}-\mathbf{YP}^t\|_F^2$	68.69 ± 8.47	72.28 ± 15.52	59.93 ± 1.96	93.03 ± 2.69	42.48 ± 3.26	80.16 ± 3.06	41.79 ± 5.13	64.17 ± 2.65	67.09 ± 3.46	18.60 ± 1.68	60.82 ± 4.79
	w/o Both	69.95 ± 8.23	73.67 ± 14.14	59.61 ± 1.59	92.83 ± 1.79	44.25 ± 4.51	80.73 ± 4.19	45.13 ± 2.76	63.72 ± 2.54	65.21 ± 2.00	18.76 ± 1.42	61.39 ± 4.32
	TEAR	**76.75 ± 1.48**	**92.16 ± 0.71**	**94.14 ± 0.66**	**94.80 ± 0.20**	**66.99 ± 1.43**	**85.10 ± 2.73**	**65.62 ± 3.16**	**75.29 ± 1.73**	**90.30 ± 2.22**	**19.53 ± 0.63**	**76.07 ± 1.49**

TABLE S6

AVERAGE RANKS OF THE COMPARED METHODS. LOWER RANKS INDICATE BETTER PERFORMANCE.

Metric	PTA-AL	PTA-CL	PTGP	SEC	GoT	FCE	YACHT	FSEC	RANGE	DREAM	TEAR
ACC	5.80	9.40	4.70	8.70	6.40	9.20	6.00	5.80	5.20	3.80	1.00
Kappa	5.80	10.00	4.40	9.50	6.50	8.80	5.70	5.50	4.70	4.10	1.00
NMI	5.00	9.90	4.10	9.20	6.10	9.80	6.10	5.50	4.00	5.20	1.10
ARI	5.60	10.10	4.30	9.50	6.30	8.90	6.00	5.40	4.20	4.50	1.20
PUR	6.00	10.40	5.10	8.80	6.20	8.60	5.80	5.60	3.95	4.35	1.20
F-score	5.50	9.00	4.30	8.90	6.40	9.80	6.00	5.80	4.60	4.30	1.40

quality consensus partitions.

D. Ablation Study

We further report ablation results in terms of ARI, PUR, and F-score in Table S5. TEAR achieves the best results on all ten datasets under ARI and PUR. Its average ARI and PUR are 66.68% and 83.11%, respectively, compared with 63.48% and 81.51% without granulation, and 38.83% and 73.09% without high-order self-expression. The average F-score is 76.07% for TEAR, compared with 74.04% without granulation and 60.82% without high-order self-expression. Removing granulation reduces the average ARI, PUR, and F-score by 3.20, 1.60, and 2.03 percentage points, respectively, while removing high-order self-expression reduces them by 27.85, 10.02, and 15.25 percentage points. Removing both yields averages of 40.51%, 74.00%, and 61.39%, respectively. These results support the contribution and complementarity of the two components.

E. Statistical Significance Analysis

We conduct Friedman tests [S1] to compare the 11 methods with complete results on all ten datasets using ACC, Kappa, NMI, ARI, PUR, and F-score. Under the null hypothesis, the methods have equivalent performance. For each dataset and

TABLE S7
FRIEDMAN TEST RESULTS WITH IMAN–DAVENPORT CORRECTION.
p_{Holm} DENOTES THE HOLM-ADJUSTED p-VALUE.

Metric	F	p	p_{Holm}
ACC	11.412	2.20×10^{-12}	4.40×10^{-12}
Kappa	15.493	9.70×10^{-16}	4.85×10^{-15}
NMI	18.064	1.34×10^{-17}	8.02×10^{-17}
ARI	15.444	1.06×10^{-15}	4.85×10^{-15}
PUR	13.298	5.29×10^{-14}	1.59×10^{-13}
F-score	11.287	2.85×10^{-12}	4.40×10^{-12}

metric, methods are ranked by their mean scores over ten runs, with rank 1 assigned to the highest score and average ranks assigned to ties. The average rank of method i is

$$\bar{r}_i = \frac{1}{N} \sum_{j=1}^{N} r_{ij},$$

where r_{ij} denotes its rank on dataset j and N is the number of datasets. Table S6 reports the average ranks; lower values indicate better performance. TEAR achieves average ranks of 1.00, 1.00, 1.10, 1.20, 1.20, and 1.40 for ACC, Kappa, NMI, ARI, PUR, and F-score, respectively. Its ranks of 1.00 for ACC and Kappa indicate first place on all ten datasets for these metrics. The consistently low ranks across the other metrics complement the detailed performance results.

The Friedman statistic is corrected for tied ranks. Let K be the number of methods and t_{ju} the number of methods in the u-th tied group on dataset j. The corrected statistic is

$$\chi_F^2 = \frac{1}{C} \left[\frac{12N}{K(K+1)} \sum_{i=1}^{K} \bar{r}_i^2 - 3N(K+1) \right],$$

where the tie correction factor is

$$C = 1 - \frac{\sum_{j=1}^{N} \sum_{u} (t_{ju}^3 - t_{ju})}{N(K^3 - K)}.$$

The Iman–Davenport F statistic is

$$F_F = \frac{(N-1)\chi_F^2}{N(K-1) - \chi_F^2},$$

where $K = 11$ and $N = 10$, yielding degrees of freedom $(10, 90)$. The six p-values are adjusted jointly using the Holm procedure.

As shown in Table S7, the Holm-adjusted p-values for all six metrics are below 0.05. Thus, the null hypothesis is rejected for each metric, indicating overall performance differences among the compared methods. These omnibus tests do not establish statistically significant pairwise differences between TEAR and every baseline; their results should be interpreted together with the average ranks and detailed clustering results.

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